

Classical Magnetic Systems

On the behaviour of macroscopic magnetized moments in dissipative
environments

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Preface

This document contains our work regarding the novel spin revolution effect, first described by PD Dr. Elena Vedmedenko and Prof. Roland Wiesen-danger[VW22]. The goal of this project is to further the understanding of the presented systems by investigating certain mathematical aspects of the system's description. Our hope is to procure either analytically solvable equations of motions or prove that such solutions do, in fact, not exist. We start by introducing the necessary mathematical principles rooted in the theory of manifolds, variational calculus and ordinary differential equations. From there on, we investigate various realizations of the spin revolution effect.
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CHAPTER ONE

Mathematical Introduction

We start with the necessary mathematical theory. If the reader already feels comfortable with the theory of smooth manifolds, variational calculus on aforementioned spaces, (Lie) groups as well as autonomous differential equations, they may skip this chapter. We assume basic knowledge of linear algebra and calculus.

1.1 Group Theory

We need some fundamental group theory. From here on, all vector spaces are finite-dimensional.

Definition 1.1 (Group). A **Group** is a pair $(G, \circ)$ consisting of a set G and an operation $\circ$ such that the following axioms are satisfied:

$$\forall a, b, c \in G : a \circ (b \circ c) = (a \circ b) \circ c$$

- $\forall a \in G \exists a^{-1} \in G : a \circ a^{-1} = 1$
- $\exists e \in G \forall a \in G : e \circ a = a$

If the operation is commutative, we call the group **abelian**.

One should think of groups in terms of symmetries: A symmetry of an abstract object can be thought of as a collection of operations leaving said object invariant.

There are plenty of examples of groups important in physics:

Example 1.2. Let V be a real vector space. The automorphisms $V \rightarrow V$ constitute a group under composition, the *general linear group* of dimension n , $GL_n(V)$. There are several important subgroups of $GL(\mathbb{R}^n)$ we will heavily use later. Let

$$\langle \cdot, \cdot \rangle : \mathbb{R}^n \rightarrow \mathbb{R}$$

be the euclidean scalar product on $\mathbb{R}^n$ and $T \in GL(\mathbb{R}^n)$. We define the *orthogonal group* as

$$O(n) := \{Q \in GL(\mathbb{R}^n) \mid QQ^t = Q^T Q = \text{id}\} \leq GL(\mathbb{R}^n).$$

Further restricting our attention to orthogonal automorphisms with unit determinant yields the *special orthogonal group*

$$SO(n) := \{R \in O(n) \mid \det R = 1\} \leq O(n)..$$

Note that every vector space over $\mathbb{R}$ has a basis. Choosing one and representing automorphisms as $n \times n$ -matrices yields the usual group structure by matrix multiplication. This also clearly demonstrates that groups of linear maps cannot be abelian in general.

In physics, we are usually concerned with how a certain group acts on a physical system. For that, we need actions and representations.

Definition 1.3 (Group Action). Let G be a group with identity e and M be a set. A **right-action** of G on M is a map

$$\alpha : G \times X \rightarrow X.$$

such that:

- $\alpha(e, x) = x$
- $\alpha(h, \alpha(g, x)) = \alpha(gh, x)$

is satisfied. We write $G \curvearrowright M$ and $\alpha(g, x) =: g.x$ for short.

Definition 1.4 (Representation). Let V be a real vector space and G be a group. A **real G -representation** is a group homomorphism

$$\rho : G \rightarrow \text{GL}(V).$$

We call V **representation space** and $\deg V$ the degree of the representation.

Apart from that, we will need the notion of an algebra, which equips vector spaces with additional structure.

Again, we do not demand commutativity. In contrast to groups, we do not demand associativity nor the existence of an identity either. Hence, associative resp. commutative algebras are explicitly denoted. An algebra with identity is called **unital**.

Definition 1.5 (Algebra). An **$\mathbb{R}$ -algebra** is a pair $(V, [\cdot, \cdot])$ where V is a real vector space and

$$[\cdot, \cdot] : V \times V \rightarrow V$$

is a *bilinear operation*, i.e. for all $u, v, w \in V$ and all $\lambda, \mu \in \mathbb{R}$ the following holds:

- Left distributivity: $[(u + v), w] = [u, w] + [v, w]$
- Right distributivity: $[u, (v + w)] = [u, v] + [u, w]$
- Scalar Compatibility: $[\lambda v, \mu w] = (\lambda \mu)[v, w]$.

While this may seem abstract at first, there are plenty of important examples in physics.

Example 1.6. We start with some easy examples from physics. The first algebra encountered in physics is probably the vector space $\mathbb{R}^3$ with the vector cross product $\times$. This is also a non-commutative, non-associative, non-unital algebra, which in turn satisfies the *Jacobi identity*

$$(u \times (v \times w)) + (v \times (w \times u)) + (w \times (u \times v)) = 0.$$

Example 1.7. In maths, the first example is usually the division algebra of the complex field $\mathbb{C}$ with multiplication of complex numbers. It is commutative, associative and unital. The term *division algebra* arises from the fact that multiplicative inverses but no zero divisors exist. Higher-dimensional variants of the complex numbers are still algebras, but less well-behaved ones. The *quaternions* $\mathbb{H}$ are still an associative and unital division algebra, but not commutative. The octonions $\mathbb{O}$ also lose associativity.

1.2 Differential Geometry

Smooth Manifolds and Tangent Spaces

From now on, we use the Einstein summation convention.

Definition 1.8 (Topological Manifold). A **topological n -manifold** is a topological space M such that:

- M has the Hausdorff property.
- M is second-countable.
- M is locally euclidean: For every $p \in M$ there is an open neighbourhood $U \subseteq M$ of p and a homeomorphism $\phi : U \rightarrow V$ such that $V \subseteq \mathbb{R}^n$ is open. We call (ϕ, U) a **chart** on M .

Since we want to apply the formalism in concrete examples, we will often work in local coordinates. Note that with the standard basis of $\mathbb{R}^n$, we can write a chart map as

$$\phi(p) = (\phi_1(p), \dots, \phi_n(p)) =: (x^1(p), \dots, x^n(p)).$$

where $x^i : M \rightarrow \mathbb{R}$ are the chart components. Since the charts are homeomorphisms, they can be locally inverted to a map $\phi^{-1} : V \rightarrow U$, in which case we call ϕ^{-1} a *local parametrization* of M .

Example 1.9. • Trivially, the real euclidean space $\mathbb{R}^n$ is an n -dimensional smooth manifold with the identity id being a global chart.

- The n -sphere $\mathbb{S}^n$ is a smooth manifold with local charts given by projection onto the coordinate axes.
- The n -torus

$$T^n = \underbrace{\mathbb{S} \times \dots \times \mathbb{S}}_{n \text{ times}}$$

is a smooth n -manifold. In the case of T^2 , one obtains local charts easily as the 2-torus is a surface of revolution.

We need the notion of smoothness on manifolds.

Definition 1.10 (Smooth Manifold). A **maximal atlas** for a topological manifold M is a collection $\mathfrak{A}$ of charts of M such that every $p \in M$ is contained in some chart and $\mathfrak{A}$ is not properly contained in some other atlas. We call $\mathfrak{A}$ smooth if for any two charts $(U, \phi), (V, \psi) \in \mathfrak{A}$ we have either $U \cap V = \emptyset$ or

$$\phi^{-1} \circ \psi : V \rightarrow U.$$

is a smooth ($\mathcal{C}^\infty$) map, called **chart transition map**. A **smooth manifold** is a pair $(M, \mathfrak{A})$ such that M is a topological manifold and $\mathfrak{A}$ is a maximal smooth atlas.

Note that with this, we get a notion of smooth maps on abstract manifolds: If M, N are smooth manifolds and $F : M \rightarrow N$ is a map, we call it smooth if for every $p \in M$ there is a chart (U, ϕ) with $p \in U$ and a smooth chart (V, ψ) with $F(p) \in V$ such that

$$\psi \circ F \circ \phi^{-1} : \mathbb{R}^m \rightarrow \mathbb{R}^n$$

is smooth in the usual sense.

There are many examples important to physics:

Example 1.11. • The $\mathbb{R}^n$ obtains a smooth maximal atlas by means of the identity.

- The sphere $\mathbb{S}^2$ has a smooth atlas with two charts $\mathbb{S}^2 \setminus N$ and $\mathbb{S}^2 \setminus S$, where N and S are north and south pole, respectively. The chart map is given by the *stereographic projection* $\sigma : \mathbb{S}^2 \setminus N \rightarrow \mathbb{R}^2$.
- Any finite-dimensional real normed vector space V is a smooth manifold. The choice of a basis determines an isomorphism $\mathbb{R}^n \cong V$ which we take as global chart. For any other basis, one obtains a basis transformation matrix which is linear, hence a $\mathcal{C}^\infty$ chart transition.
- Any open subset $U \subseteq \mathbb{R}^n$ is a smooth manifold on its own with the smooth atlas $\{U, \text{id}_U\}$. We call such a manifold an *open submanifold* of $\mathbb{R}^n$.

With smooth maps on manifolds, we are already able to talk about curves or paths on such spaces. However, in physics one is especially interested in velocity and acceleration. For this, we will need two additional structures: The tangent bundle, which comes naturally with every smooth manifold, and a metric to measure distances and angles. We define a **curve** on a manifold M to be a smooth map

$$\gamma : I \rightarrow M$$

where $I \subseteq \mathbb{R}$ is some interval.

Definition 1.12 (Tangent Bundle). Let M be a smooth manifold and $p \in M$. We declare two smooth curves $\gamma_i : I \rightarrow M$, $i \in \{1, 2\}$ to be equivalent if for any chart (U, ϕ) of p we have

$$\left. \frac{d}{dt}(\phi \circ \gamma_1) \right|_{t=0} = \left. \frac{d}{dt}(\phi \circ \gamma_2) \right|_{t=0}. \quad (1.1)$$

The space of equivalence classes of such curves is called **tangent space** at p and denoted $T_p M$. The **tangent bundle** of M is the disjoint union

$$TM := \coprod_{p \in M} T_p M.$$

Note that while the tangent space at a point on an n -manifold is an n -dimensional real vector space (and hence isomorphic to a copy of $\mathbb{R}^n$), the tangent bundle is a $2n$ -dimensional space and indeed also a smooth manifold. However, it is not always possible to identify $TM \cong M \times \mathbb{R}^n$. A point in TM is a pair (p, v) , where $p \in M$ and $v \in T_p M$. We also have a natural projection $\pi : TM \rightarrow M$, given by $\pi(p, v) = p$.

Tangent vectors act on smooth functions $f : M \rightarrow \mathbb{R}$ by

$$v(f) = \left. \frac{d}{dt} \right|_{t=0} (f \circ \gamma)(t).$$

Note that this gives rise to the usual partial derivative of a function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ by setting

$$\left. \frac{\partial g}{\partial x^i} \right|_y := \frac{d}{dt} g(y + te_i).$$

where γ represents one path with $\dot{\gamma}|_{t=0} = v$. Using the fact that for every $p \in M$ there is a chart $(U, (x^1, \dots, x^n))$ centered around p , we are able to pull back the basis of $T_{\phi(p)} \mathbb{R}^n$ to $T_p M$. The basis of $T_{\phi(p)} \mathbb{R}^n$ is given by paths of the form $t \mapsto \phi(p) + te_i$, where e_i is the i -th canonical basis vector of $\mathbb{R}^n$. Writing this basis as $(\partial_{x^1}, \dots, \partial_{x^n})$, we can pull back along ϕ^{-1} to obtain a basis of $T_p M$, defined as

$$\partial_i|_p := \phi^{-1}(\phi(p) + te_i).$$

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This expression acts on a smooth function as above by

$$\partial_i|_p f = \left. \frac{\partial(f \circ \phi^{-1})}{\partial x^i} \right|_{\phi(p)}.$$

Definition 1.13 (Differential). Let $F : M \rightarrow N$ be a smooth map between smooth manifolds. The **differential** of F at $p \in M$ is a linear map

$$dF_p : T_p M \rightarrow T_{F(p)} N$$

given by

$$dF_p([\gamma]) = [F \circ \gamma].$$

where $[\gamma]$ is an equivalence class of curves through p .

Given local charts $(U, (x^1, \dots, x^m))$ around p and $(V, (y^1, \dots, y^n))$ around $F(p)$, the differential can be expressed in local coordinates as

$$dF_p(\partial_i) = \partial_i F^j|_p \hat{\partial}_j|_{F(p)}.$$

with $F^j = y^j \circ F$ and $\hat{\partial}_j$ being a basis vector of $T_{F(p)} N$. We conclude this section with the definition of smooth vector fields.

Definition 1.14 (Vector Fields). Let M be a smooth manifold. A **smooth vector field** is a smooth section

$$V : M \rightarrow TM$$

such that $\pi \circ V = \text{id}_M$ with the natural projection $\pi : TM \rightarrow M$. The space of smooth sections on M is denoted $\mathfrak{X}(M)$.

Once more, we desire a local coordinate expression. If $(U, (x^1, \dots, x^n))$ is a local chart, we can use the basis of $T_p M$ defined above for all $p \in U$ to write

$$X_p = X^i(p) \partial_i|_p$$

where $X^i : M \rightarrow \mathbb{R}$ are the component functions of X . If a collection of vector fields $(X_1, \dots, X_n)$ spans $T_p M$ locally or globally, we call it a **local or global frame**. Directing our attention one last time to curves, we can now define curves whose velocity vectors form a given vector field.

Definition 1.15 (Integral Curves). Let M be a smooth manifold and $V \in \mathfrak{X}(M)$. An **integral curve** of V is a smooth curve $\gamma : I \rightarrow M$ such that

$$\frac{d}{dt} \gamma(t) = V_{\gamma(t)}$$

for all $t \in I$.

We need this notion of walking along the path prescribed by a vector field later for Hamiltonian phase flows. In general, a flow is the rigorous version of this intuitive idea of following a field:

Note that vector fields also act naturally on smooth functions $f : M \rightarrow \mathbb{R}$ by means of partial derivation:

$$(Xf)(p) = X_p f = X^i(p) \partial_i|_p(f).$$

Intuitively, one should think of $t \in \mathbb{R}$ as time. A global flow then shifts the whole manifold along some vector field, e.g. the torus rotates under the global flow of a radial vector field.

Definition 1.16 (Flow). Suppose M is a smooth manifold with $V \in \mathfrak{X}(M)$ such that a unique global integral curve of V exists for all $p \in M$. Denoting the integral curve at p by $\gamma_p : \mathbb{R} \rightarrow M$, a **global flow** is a continuous action of $\mathbb{R}$ on M :

$$\begin{aligned}\theta : M \times \mathbb{R} &\rightarrow M \\ (p, t) &\mapsto \gamma_p(t).\end{aligned}$$

Riemannian Metrics

To measure distances and angles, we need additional structure on our smooth manifolds. Dual to the tangent bundle of M , we introduce the **cotangent bundle**

$$T^*M := \coprod_{p \in M} (T_p M)^*$$

which is, as usual in linear algebra, given by the space of linear forms

$$\omega : TM \rightarrow \mathbb{R}.$$

The basis dual to $(\partial_1, \dots, \partial_n)$ will be denoted $(dx^1, \dots, dx^n)$.

Definition 1.17 (Riemannian Manifolds). A **Riemannian metric** on a smooth manifold M is a smooth, symmetric and positive-definite bilinear form

$$g : TM \times TM \rightarrow \mathbb{R}.$$

The pair (M, g) is called **Riemannian manifold**.

This also yields the usual notions obtained with a scalar product on $\mathbb{R}^n$: We have a norm on each $T_p M$ defined by

$$\|v\|_g := \sqrt{\langle v, v \rangle}$$

and an angle given by

$$\cos \angle(v, w) := \frac{\langle v, w \rangle}{\|v\|_g \|w\|_g} \in [0, \pi].$$

Since each g_p is non-degenerate, we even have the well-known musical isomorphisms $T_p M \cong T_p^* M$. If $X = X^i \partial_i$ is a vector field, we identify it with $X^\flat = g_{ij} X^i dx^j$. A covector field $\omega = \omega_i dx^i$ is identified with $\omega^\sharp = g^{ij} \omega_j \partial_i$ where $g^{ij} = g_{ij}^{-1}$.

In local coordinates at some $p \in M$, we can use the canonical basis of $T_p M$ to write $g_p : T_p M \times T_p M \rightarrow \mathbb{R}$ as

$$g_p(v, w) = g_p(v^i \partial_i|_p, w^j \partial_j|_p) =: \langle v^i \partial_i|_p, w^j \partial_j|_p \rangle = v^i w^j g_{ij}$$

where $g_{ij} = g_p(\partial_i|_p, \partial_j|_p)$ denotes the matrix representation of the metric g . The most important definition arising from the existence of a metric function is in our case the arc length.

Definition 1.18 (Length). Let (M, g) be a smooth Riemannian manifold and $\gamma : I \rightarrow M$ be a smooth curve. The **length** of γ is given by the functional

$$L[\gamma] := \int_I \|\partial_t \gamma(t)\|_g dt..$$

We will now connect those concepts with physics' formalism.

1.3 Lie Groups and Lie Algebras

1.3.1 Definitions and Properties

The preceding chapters dealt with the category of groups and the category of smooth manifolds, respectively. Those two categories can be combined to

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form a new category which will turn out to be important for the rotational configuration space of a rigid body. After a gentle introduction, we continue by treating the Lie group most important to us: $SO(3)$.

Definition 1.19 (Lie Group). A **smooth Lie group** is a triple $(M, \circ, \mathfrak{A})$ such that $(M, \circ)$ is a group, $(M, \mathfrak{A})$ is a smooth manifold with atlas $\mathfrak{A}$, and

$$(g, h) \mapsto g \circ h$$

as well as

$$g \mapsto g^{-1}$$

are smooth maps.

Example 1.20. The most important examples to physicists are certainly matrix Lie groups. We already mentioned the real *general linear group* $GL_n(\mathbb{R})$, which is also a Lie group. So are the (Lie) subgroups $SO(n)$ and $O(n)$. It is expected that groups of linear maps have smooth multiplication and inversion as linear maps are smooth in general. However, this can only be made rigorous by topologizing GL_n with the metric topology of $\mathbb{R}^{n^2}$ element-wise. All mentioned subgroups then inherit the subspace topology, yielding a Lie group structure.

The main reason for the importance of Lie groups is their connection to symmetry. To understand this, we let some Lie group G act on another smooth manifold M . Expanding our definition of action, we call such an action

$$\rho : G \times M \rightarrow M$$

smooth if for all $g \in G$ and $x \in M$, the map

$$\rho_g : M \rightarrow M, x \mapsto \rho_g(x) = g.x$$

is a smooth map between manifolds.

Our notion of symmetry arises from this definition, but we keep it somewhat informal. A **symmetry** of a manifold M with some structure is the action of some Lie group on M preserving the structure.

Example 1.21. One of the most striking examples of the importance of Lie groups is the fact that different flavours of geometry arise from Lie group actions. Consider the Euclidean space $\mathbb{R}^n$ as Lie Group: Obviously, it is a smooth manifold by definition, it just remains to choose addition as group operation to obtain a Lie group. Then, the **Euclidean group**

$$E(n) := \mathbb{R}^n \rtimes O(n)$$

acts on $\mathbb{R}^n$ by

$$\rho_{\tau, A}(v) := \tau + A(v)$$

where $\tau, v \in \mathbb{R}^n$ and $A \in O(n)$. The multiplication inside $E(n)$ is given by the usual semidirect product multiplication:

$$(\tau, A) \circ (\tau', A') = (\tau + A(\tau'), AA').$$

One example for such a structure would, of course, be a metric. If an action ρ preserves a metric g on M , we can write this explicitly as

$$g(\rho_h(x), \rho_h(y)) = g(h.x, h.y) = g(x, y)$$

for all $h \in G$ and $x, y \in M$.

This semidirect product group may seem confusing at first, but this case can be made very intuitive by thinking about a physical system in $\mathbb{R}^n$ with fixed time (for now) and considering only $SO(n) \leq O(n)$. Then, the action of $\mathbb{R}^n \rtimes SO(n)$ first rotates v by the rotation matrix R and then translates the rotated frame by τ . This also demonstrates why the interior group multiplication has to have this particular form: If one applies first a rotation R' and then a translation τ' , followed by the same procedure with (R, τ) , the second rotation R has to be done from the perspective of the rotated frame, so we rotate τ' by R before adding the second translation τ . Considering $E(n)$ again, we allow all transformations preserving orthogonality (e.g. reflection), but the same idea applies.

This action is at the core of Euclidean geometry, meaning that Euclidean geometry studies objects invariant under the action of $E(n)$. If one replaces $O(n)$ for example by the whole of GL_n , we arrive at affine geometry. More physically motivated examples would be the action of the Lorentz group on Minkowski space-time (modeled as $\mathbb{R}^4$ with the Minkowski metric) or the action of the symplectic group on symplectic manifolds, which govern Hamiltonian mechanics.

With this idea in mind, we take a look at representations again. Of special interest are so-called *faithful* representations $\varrho : G \rightarrow GL_n$ of some Lie group G which are *injective*.

Example 1.22. We continue with the study of $E(n)$.

Luckily, matrix Lie groups are already subgroups of GL_n , so we can just take the inclusion as representation. The representation of $E(n)$ needs to take care of one additional translation, which is not represented by $GL(n)$. We obtain a map $\varrho : E(n) \rightarrow GL_n$ which sends $(\tau, A) \in E(n)$ to

$$\varrho(\tau, A) := \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \in GL(n+1, \mathbb{R}).$$

Lie groups do not only interact well with manifolds, but are also interesting by themselves. We want to shortly explore the tangent space of a Lie group. Since every Lie group has a smooth multiplication, we always have an action

$$L_h : G \times G \rightarrow G, g \mapsto L_h(g) = h \circ g$$

for some fixed $h \in G$. This is called **left translation**. It gives rise to a map

$$(dL_h)_g : T_g G \rightarrow T_g G$$

via the usual differential. Now, we consider smooth vector fields on our Lie group and give them more structure.

Note that the Lie bracket itself can be defined for arbitrary smooth manifolds, not only Lie groups. Also, we have an expression in local coordinates as

$$[X, Y] = (X^i \partial_i Y^j - Y^i \partial_i X^j) \partial_j.$$

Definition 1.23 (Lie Bracket). Let M be a smooth manifold and $X, Y \in \mathfrak{X}(M)$. The **Lie bracket** is an operator

$$[\cdot, \cdot] : \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M),$$

mapping a smooth function $f : M \rightarrow \mathbb{R}$ to another smooth function defined by

$$[X, Y]f = XY(f) - YX(f).$$

We can directly see that for left invariant vector fields X, Y , the Lie bracket is also left invariant. Hence, the vector space of left invariant vector spaces is closed under one additional operation. This hints at another structure we already touched upon.

Definition 1.24 (Lie Algebra). A **Lie Algebra** over $\mathbb{R}$ is an algebra $(\mathfrak{G}, [\cdot, \cdot])$ such that two additional identities hold:

- Antisymmetry: $[X, Y] = -[Y, X]$
- Jacobi Identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

Example 1.25. The most important example is the one we just constructed. Given a Lie group G , the space of smooth, left invariant vector fields on L , denoted $\mathfrak{g}$, is a Lie algebra with the Lie bracket. Separated from Lie groups, one can consider smooth vector fields on a smooth manifold in general. Another example of a Lie algebra is $\mathbb{R}^3$ together with the vector cross product, which we already encountered and will encounter again.

Now, we conclude the general discussion with a theorem, yielding fundamental insight into the relationship of a Lie group and its associated algebra.

Theorem 1.26 (Lie Group-Algebra Correspondence). Let G be a Lie group with identity $\text{id} \in G$ and let $\mathfrak{g}$ be the Lie algebra of left-invariant vector fields on G . Then the evaluation

$$\mathfrak{g} \ni X \mapsto X|_{\text{id}} \in T_{\text{id}}G$$

of a left-invariant vector field at the identity is a vector space isomorphism $T_{\text{id}}G \cong \mathfrak{g}$.

This means that the best linear approximation of a Lie group at the identity is its Lie algebra. In lieu of group theory, one calls Lie algebras linearized Lie groups.

1.3.2 The Special Orthogonal Group

Example 1.27. Bridging to our discussion of $\text{SO}(3)$, we demonstrate how to calculate the Lie algebra $\mathfrak{so}(3)$. By the preceding theorem, the Lie algebra is isomorphic to a tangent vector at the identity matrix $\text{id} \in \text{SO}(3)$. For now, we avoid parametrizing elements of $\text{SO}(3)$ and utilize the Taylor expansion for manifolds. Hence, choose a smooth path

$$(-\epsilon, \epsilon) \ni t \mapsto A(t) \in \text{SO}(3)$$

with $A(0) = \text{id}$. We can expand up to first order as

$$A(t) = \text{id} + \dot{A}|_{t=0} t + \mathcal{O}(t^2).$$

Since $A(t)$ is in $\text{SO}(3)$ for all t , we demand $A(t)A^\top(t) = \text{id}$. Writing $\dot{A} \equiv \dot{A}|_{t=0}$ and plugging in, we obtain

$$\text{id} = A(t)A^\top(t) = (\text{id} + \dot{A} + \mathcal{O}(t^2))(\text{id} + \dot{A} + \mathcal{O}(t^2))^\top = \text{id} + (\dot{A} + \dot{A}^\top) \cdot t + \mathcal{O}(t^2).$$

Dropping all non-linear terms, we obtain

$$\dot{A}^\top = -\dot{A}.$$

Thus, $T_{\text{id}} \text{SO}(3)$ consists of anti-symmetric 3×3 -matrices and we can write

$$\mathfrak{so}(3) = \{\Omega \in \text{Mat}_3(\mathbb{R}) \mid \Omega^\top = -\Omega\}.$$

Skew-symmetric matrices have only three degrees of freedom. Choose three real numbers $(\omega_1, \omega_2, \omega_3)$. Then, we can write a general element of $\mathfrak{so}(3)$ as

$$\Omega = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}.$$

This yields a basis for $\mathfrak{so}(3)$ in terms of three matrices:

$$(\xi_1, \xi_2, \xi_3) = \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right).$$

Considering those numbers as vector $\omega = (\omega_1, \omega_2, \omega_3) \in \mathbb{R}^3$, we obtain a Lie algebra isomorphism

$$(\mathbb{R}^3, \times) \cong (\mathfrak{so}(3), [\cdot, \cdot]),$$

where $\times$ is the vector cross product and $[A, B] = AB - BA$ is the usual matrix commutator.

We summarize this discussion in a proposition and investigate $SO(3)$ deeper.

Theorem 1.28 (Infinitesimal Rotation Algebra). The Lie algebra $\mathfrak{so}(3)$ corresponding to the Lie group $SO(3)$ is the space of skew-symmetric, real, 3×3 -matrices with basis (ξ_1, ξ_2, ξ_3) . The ξ_i are called **infinitesimal generators** of $\mathfrak{so}(3)$. There is a Lie algebra isomorphism

$$\mathfrak{so}(3) \cong \mathbb{R}^3,$$

mapping $\mathbb{R}^3 \ni \omega \mapsto \sum \omega_i \xi_i \in \mathfrak{so}(3)$.

Purposefully, we already chose notations fitting with the physical interpretation: The vector $\omega \in \mathbb{R}^3$ is an *axis-angle vector*, where $\varpi := \frac{\omega}{\|\omega\|}$ defines the axis of rotation and $\|\omega\|$ defines the angle of rotation. If we consider a rigid body rotating at a point fixed in space, we can assign a coordinate frame of $\mathbb{R}^3$ to that body, rotating synchronous with it. The state of that frame at any time t can then be described as a rotation of the initial system, hence as an element of $SO(3)$ acting on a vector in $\mathbb{R}^3$. The infinitesimal generators of rotation with respect to the rotating frame are then elements of $\mathfrak{so}(3)$. Additionally, if the body is free to move in space we add a stationary frame to describe the center of mass motion. The motion of the system is then a curve $t \mapsto \mathbb{R}^3 \times SO(3)$. As of yet, we are devoid of a chart for $SO(3)$, which will be needed to do practical calculations.

While $\mathfrak{so}(3)$ is remarkably easy to parametrize, $SO(3)$ is much more complicated. We begin by stating a result demonstrating the best case for charts on $SO(3)$.

Theorem 1.29 (Lusternik-Schnirelmann Theorem). Let M be a smooth compact manifold and $\text{crit}(M)$ be the minimum number of critical points a smooth function $f : M \rightarrow \mathbb{R}$ can have. Then,

$$\text{LSC}(M) \leq \text{crit}(M) - 1,$$

where LSC denotes the Lusternik-Schnirelmann category.

This may seem weirdly abstract at first, but it is well-known that $SO(3) \cong \mathbb{RP}^3$ as groups, and another result of Lusternik-Schnirelmann states that $\text{LSC}(\mathbb{RP}^3) = 5$. Hence, any smooth function on $SO(3)$ has at least four critical points. This proves the following:

Corollary 1.30 (Maximal Atlas of $SO(3)$). A maximal atlas of $SO(3)$ consists of at least four charts.

Geodesic Polar Coordinates Our first attempt to obtain such a minimal chart will concern the question how much of a Lie group can be recovered from the linearized version at the origin - the Lie algebra. We need the notion of *one-parameter subgroups*, which are just Lie group homomorphisms

$$\gamma : (\mathbb{R}, +) \rightarrow G, ,$$

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where G is some Lie group. It turns out that these subgroups are precisely maximal integral curves of left-invariant vector fields on G starting at $\text{id} \in G$.

Definition 1.31 (Lie Exponential). Let G be a Lie group with corresponding Lie algebra $\mathfrak{g}$. The map

$$\begin{aligned}\exp : \mathfrak{g} &\rightarrow G \\ X &\mapsto \exp(X) := \gamma_X(1)\end{aligned}$$

where γ_X is the one-parameter subgroup generated by X is called **(Lie) exponential**.

Fortunately, this exponential coincides with the usual matrix exponential

$$\exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

for matrix Lie groups such as $\text{SO}(3)$. Indeed, $\exp$ is smooth and non-singular at the identity. Hence, by the inverse function theorem $\exp$ is a diffeomorphism around a small enough neighbourhood of the identity (it can not be a global diffeomorphism since we know that $\mathfrak{so}(3) \cong \mathbb{R}^3 \not\cong \text{SO}(3)$).

Theorem 1.32 (Rodrigues' Theorem). Given an axis-angle rotation vector $\omega \in \mathbb{R}^3$ and setting $\theta = \|\omega\|$ as well as $\varpi = \frac{\omega}{\theta}$, we identify ω and ϖ with their corresponding skew-symmetric matrices in $\mathfrak{so}(3)$ as above. Then, the Lie exponential is given by *Rodrigues Rotation Formula*

$$\exp(\omega) = \text{id} + \frac{\sin \theta}{\theta} \omega + \frac{1 - \cos \theta}{\theta^2} \omega^2. \quad (1.2)$$

Proof. Writing an arbitrary axis-angle vector $\omega \in \mathbb{R}^3$ as $\omega = \theta \varpi$, we can use the isomorphism $\mathbb{R}^3 \cong \mathfrak{so}(3)$ to identify ϖ with its skew-symmetric matrix as above. This yields

$$\exp(\omega) = \sum_{k=0}^{\infty} \frac{(\theta \varpi)^k}{k!}.$$

Now, we calculate the characteristic polynomial of ϖ :

$$p_{\varpi}(t) = \det(\varpi - t \cdot \text{id}) = \det \begin{pmatrix} -t & -\varpi_1 & \varpi_2 \\ \varpi_1 & -t & -\varpi_3 \\ -\varpi_2 & \varpi_3 & t \end{pmatrix} = -(t^3 + t)$$

where we used that $\sum \varpi_i^2 = 0$ as ϖ is normalized. By the Cayley-Hamilton theorem, $p_{\varpi}(\varpi) = 0$. It follows that $\varpi^3 = -\varpi$. This means that the power series repeats after the third power with $\varpi^3 = -\varpi$, $\varpi^4 = -\varpi^2$ and $\varpi^5 = \varpi$, cycling like this indefinitely. Rearranging the matrix exponential, we obtain

$$\exp(\omega) = \text{id} + \left(\sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k+1}}{(2k+1)!} \right) \varpi + \left(1 - \sum_{k=1}^{\infty} (-1)^k \frac{\theta^{2k}}{(2k)!} \right) \varpi^2,$$

but this is nothing other than the trigonometric Taylor series of $\sin$ and $\cos$. Resubstituting ϖ with ω , we obtain our claim:

$$\exp(\omega) = \text{id} + \frac{\sin \theta}{\theta} \omega + \frac{1 - \cos \theta}{\theta^2} \omega^2.$$

□

With this explicit formula, we see that $\exp$ serves only as local chart, just like we predicted, since there are obvious degeneracies at $\theta = \pi$ and $\theta = 0$. We conclude with our first minimal atlas.

The name of this chart is not arbitrary: The Lie exponential coincides with the Geodesic exponential in this case. This means that exponential polar charts indeed describe the fastest path between an element of $\text{SO}(3)$ and the identity, provided one stays in the domain where $\exp$ is a local diffeomorphism.

Theorem 1.33 (Geodesic Polar Chart). The *Geodesic Polar Chart* of $\text{SO}(3)$ is the Lie exponential

$$\exp : \{\Omega \in \mathfrak{so}(3) \mid \|\Omega\| < \pi\} \rightarrow \text{SO}(3),$$

which is a proper chart with inverse

$$\log : \text{SO}(3) \setminus \{R \in \text{SO}(3) \mid \text{tr } R = -1\} \rightarrow \mathfrak{so}(3).$$

One can show that the main chart given by $\exp$ together with three maps composed of $\exp$ and a rotation around respectively one of the three spacial orientations by π suffices to obtain an atlas.

Euler Coordinates Now, we want to investigate another chart which takes a more direct approach. Knowing that $\text{SO}(3)$ represents rotations in $\mathbb{R}^3$, we can describe any rotation by a combination of three rotations around (rotated) basis vectors. This gives rise to 12 possible ways of decomposing a rotation into three basis rotations, each of the three being characterized by an angle. These angles are called Euler angles, and they yield a minimal global atlas on $\text{SO}(3)$. We will work with the $zy'z''$ -convention, first rotating around z by ϕ , then around the rotated y -axis y' by ψ , and finally around the rotated z -axis z'' by θ .

For rotation matrices, we write $\sin x \equiv s_x$ and $\cos x \equiv c_x$.

Theorem 1.34 (Euler Angle Chart). The map

$$\mathfrak{E} : T^3 \rightarrow \text{SO}(3)$$

$$(\phi, \psi, \theta) \mapsto \begin{pmatrix} c_\phi c_\psi c_\theta - s_\phi s_\psi & -c_\phi c_\psi c_\theta - s_\phi s_\psi & s_\theta c_\psi \\ s_\psi s_\theta c_\phi + s_\phi c_\psi & -s_\psi s_\theta c_\phi + s_\phi c_\psi & s_\psi s_\theta \\ -s_\theta c_\phi & s_\theta s_\phi & c_\theta \end{pmatrix}$$

restricts to four local charts covering $\text{SO}(3)$.

It is noteworthy that this is really not a covering in the sense of topological spaces: The Euler angle map has singularities at which $\mathfrak{E}$ fails to be a local homeomorphism. This is called *gimbal lock*.

Quaternion Coordinates The singularities of the Euler chart are cumbersome to deal with numerically, so we desire a chart without singularities. Also, we will see that unit quaternions are always representing a rotation. This

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means that as long as a unit quaternion is varied appropriately, we will not end up with a quaternion which is not a rotation anymore (which is obviously not the case for e.g. orthogonal matrices). The drawback will be a higher degree of redundancy compared to the other charts.

Since the Euler chart fails to be a covering space because of singular points, it seems reasonable to start with a topological covering of $SO(3)$: The universal covering space $SU(2)$. We begin by defining all relevant parts of the **quaternion algebra**, denoted $\mathbb{H}$.

The quaternions are a real 4-dimensional vector space with basis

$$\mathbf{1} := (1, 0, 0, 0)^\top, \mathbf{i} := (0, 1, 0, 0)^\top, \mathbf{j} := (0, 0, 1, 0)^\top, \mathbf{k} := (0, 0, 0, 1)^\top$$

and component-wise vector addition and scalar multiplication. We define a multiplication on $\mathbb{H}$, called **Hamilton product**, by specifying the rules on the basis and extending by distributivity. The **scalar quaternion 1** will be the two-sided identity:

$$\mathbf{1}i = i\mathbf{1} = i, \mathbf{1}j = j\mathbf{1} = j, \mathbf{1}k = k\mathbf{1} = k.$$

The **vector quaternions** obey the following laws:

$$ij = -ji = k, jk = -kj = i, ki = -ik = j$$

and

$$i^2 = j^2 = k^2 ijk = -1.$$

This turns $\mathbb{H}$ into a skew field, i.e. a field without commutativity. We have a unary operation on $\mathbb{H}$, called **conjugation**. Given a quaternion $q = a + bi + cj + dk$, define the conjugate as

$$q^* := a - (bi + cj + dk).$$

Furthermore, $\mathbb{H}$ is a normed vector space with norm given by

$$\|q\| := \sqrt{qq^*}.$$

We denote the scalar or real part by $\Re(q) := a$ and the vector or imaginary part by $\Im(q) := bi + cj + dk$. With this, we define the subspace of unit quaternions or **versors**, i.e. the 3-sphere of $\mathbb{H}$:

$$\mathbb{S}^3 := \{q \in \mathbb{H} \mid \|q\| = 1\} \subset \mathbb{H}.$$

The versors form a group with the Hamilton product. In addition, Eulers formula generalizes to $\mathbb{S}^3$:

Proposition 1.35 (Euler's Formula). Any versor $u \in \mathbb{S}^3$ can be written as

$$u = \exp(\theta \varpi) = \cos(\theta) + \varpi \sin(\theta)$$

where θ is the polar angle

$$\theta := \arccos(\Re(u)) \in [0, \pi]$$

and ϖ is an imaginary unit quaternion

$$\varpi := \frac{\Im(u)}{\|\Im(u)\|}.$$

This may seem obvious, but a desirable consequence of versors being a subgroup is that they are closed under multiplication, i.e. multiplying two versors will always produce another versor.

CHAPTER 1. MATHEMATICAL INTRODUCTION

We will now identify $\mathbb{R}^3$ with the space of pure quaternions $\Im\mathbb{H} := \{\mathbf{v} \in \mathbb{H} \mid \Re(\mathbf{v}) = 0\}$, i.e. $\mathbb{R}^3 \ni (a, b, c)^\top \leftrightarrow a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \in \Im\mathbb{H}$. It is obvious that as vector spaces, vector addition and scalar multiplication are precisely the same. The question is how the multiplicative structure on $\mathbb{H}$ relates to the cross product on $\mathbb{R}^3$. Direct calculation shows that for $\mathbf{u} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, $\mathbf{v} = d\mathbf{i} + e\mathbf{j} + f\mathbf{k} \in \mathbb{R}^3 \cong \Im\mathbb{H}$, we have

$$\mathbf{u}\mathbf{v} = (-\langle \mathbf{u}, \mathbf{v} \rangle, \mathbf{u} \times \mathbf{v}).$$

With scalar and cross products, we have enough tools to describe rotation with quaternions. In $\mathbb{R}^3$, we already saw that rotations are parametrized by an angle of rotation θ around a rotational axis unit vector $\boldsymbol{\varpi}$. Under the correspondence, $\boldsymbol{\varpi} = \varpi_1\mathbf{i} + \varpi_2\mathbf{j} + \varpi_3\mathbf{k}$. We define a unit quaternion

$$\mathbf{u} := \exp\left(\frac{\theta}{2}\boldsymbol{\varpi}\right) = \cos\frac{\theta}{2} + \boldsymbol{\varpi}\sin\frac{\theta}{2}.$$

The reader may verify that $\boldsymbol{\varpi}$ interpreted as a pure quaternion is indeed also a unit quaternion.

Define the **linear conjugation transformation**

$$\begin{aligned} L : \Im\mathbb{H} &\rightarrow \Im\mathbb{H} \\ \mathbf{p} &\mapsto L_{\mathbf{u}}(\mathbf{p}) := \mathbf{u}\mathbf{p}\mathbf{u}^{-1} \end{aligned}$$

for some $\mathbf{q} \in \mathbb{S}^3$. By calculation, one can show that an arbitrary vector $\mathbf{p} \in \mathbb{R}^3$ is mapped to $L_{\mathbf{u}}(\mathbf{p})$ under the rotation represented by $\mathbf{u}$. Hence any rotation can be represented as a unit quaternion. However, a rotation about $\boldsymbol{\varpi}$ by θ and a rotation about $-\boldsymbol{\varpi}$ by $2\pi - \theta$ produces the same physical result (just with a different path). In versor representation, if the positive rotation corresponds to $\mathbf{u}$, the negative one corresponds to $-\mathbf{u}$, but $\mathbf{u} \neq -\mathbf{u}$ in $\mathbb{H}$. This means that the map $\mathbb{S} \rightarrow \text{SO}(3)$ fails to be injective globally, it is a double cover.

Theorem 1.36 (Quaternion Coordinates). The map

$$\begin{aligned} \Omega : \mathbb{S}^3 &\rightarrow \text{SO}(3) \\ \mathbf{q} &\mapsto \begin{pmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1q_2 - q_3q_0) & 2(q_1q_3 + q_2q_0) \\ 2(q_1q_2 + q_3q_0) & 1 - 2(q_1^2 + q_3^2) & 2(q_2q_3 - q_1q_0) \\ 2(q_1q_3 - q_2q_0) & 2(q_2q_3 + q_1q_0) & 1 - 2(q_1^2 + q_2^2) \end{pmatrix} \end{aligned}$$

is the universal covering space of $\text{SO}(3)$, a double cover, and restricts to local charts.

Lastly, we also want local conversion formula between quaternions and other charts:

Lemma 1.37 (Axis-Angle Conversion). Given a versor $\mathbf{u} \in \mathbb{S}^3$, the axis of rotation is given by

$$\boldsymbol{\varpi} = \frac{\Im(\mathbf{u})}{\|\Im(\mathbf{u})\|}$$

and the angle of rotation is given by

$$\theta = 2 \arctan2(\|\Im(\mathbf{u})\|, \Re(\mathbf{u})).$$

Given an axis-angle rotation $(\theta, \boldsymbol{\varpi})$, we have already seen that the representing quaternion is

$$\mathbf{u} = \exp\left(\frac{\theta}{2} \boldsymbol{\varpi}\right).$$

We denote by $\arctan2(x, y)$ the two-argument arc-tangent which returns the angle between the ray through (x, y) and the x -axis, i.e. the argument of $y + ix \in \mathbb{C}$.

Lemma 1.38 (Rotational Matrix Conversion). Given a versor, we already saw the covering map restricting to local charts in Theorem 1.36. Given a rotational matrix R , we can diagonalize it to obtain the eigenvector $\boldsymbol{\varpi}$ of the eigenvalue 1, which is the axis of rotation. The angle is given by $\theta = \arcsin\left(\frac{\text{tr}(R)-1}{2}\right)$. With that, we reduced to the axis-angle conversion already solved above.

The last noteworthy formula concerns the carry-over of vector-valued derivatives in $\mathbb{R}^3$ with the cross product. Given a vector $\mathbf{p} \in \mathbb{R}^3 \cong \Im\mathbb{H}$ and a rotation $\mathbf{q} \in \mathbb{S}^3$, we have

$$\frac{dL_{\mathbf{q}}(\mathbf{p})}{d\mathbf{q}} = \mathbf{qp} - (\mathbf{qp})^* + [(\mathbf{qp}i)^* - \mathbf{qp}i]i + [(\mathbf{qp}j)^* - \mathbf{qp}j]j + [(\mathbf{qp}k)^* - \mathbf{qp}k]k.$$

1.4 Lagrangian Mechanics

1.4.1 Calculus of Variations

The central concept is concerned with a special function, the *Lagrangian*. While its definition may seem abstract and purely mathematical, it will turn out to be central to a formulation of classical physics on par with Newton's axioms.

Definition 1.39 (Lagrangian). Let M be a smooth manifold with tangent bundle TM . A **Lagrangian** is a smooth function

$$\mathcal{L} : TM \times \mathbb{R} \rightarrow \mathbb{R}.$$

If $\mathcal{L}$ is constant in the second argument, we call it **autonomous**.

A priori, we do not impose any given form on $\mathcal{L}$. If we choose local coordinates $\Phi = (x^1, \dots, x^n, \partial_1, \dots, \partial_n)$, the Lagrangian can be expressed as

$$\mathcal{L}(x^1, \dots, x^n, \partial_1, \dots, \partial_n, t) := \mathcal{L} \circ \Phi^{-1} : \mathbb{R}^{2n} \times \mathbb{R} \rightarrow \mathbb{R}.$$

Bridging between smooth manifolds and physical systems is the concept of

configuration manifolds and (velocity) phase spaces as well as the theorem of Euler-Lagrange.

To be consistent with notational conventions in physics, we will choose the letter q for coordinates on configuration manifolds. Hence, local coordinates are now given as $\phi = (q^1, \dots, q^n)$.

Definition 1.40 (Configuration Manifold). The **configuration manifold** Q of a system of N point masses consists of all possible positions of those N masses in Euclidean 3-space. We call $\dim Q$ the **degrees of freedom**.

With this definition, which will suffice for the systems considered later, all configuration manifolds are some subsets of $\mathbb{R}^{3N}$. However, we aim at describing configuration manifolds more precisely than that.

Example 1.41. • The configuration manifold of N free point masses in Euclidean 3-space is given by $\mathbb{R}^{3N}$ as each particle is free to move in three spacial directions independently.

- The two-dimensional double pendulum has the configuration manifold $Q = \mathbb{S}^1 \times \mathbb{S}^1$ as each point can move in one copy of $\mathbb{S}^1$. Note that $\mathbb{S}^1 \times \mathbb{S}^1 \cong T$, so we can think of Q as a 2-torus.
- A rigid rod in Euclidean 2-space is free to rotate around an axis at its end. The configuration manifold is $Q = \mathbb{R}^2 \times \mathbb{S}^1$ since we need two coordinates (q^1, q^2) to describe the position of the axis of rotation and one angle $q^3 \in [0, 2\pi]$ to describe the rotation itself.

From classical physics we know very well that positions are not sufficient to characterize a system fully, one also needs velocities.

To give a quick overview: We have $\dim TQ = 2 \cdot \dim Q$, so we need twice as much coordinates on TQ . Those are given by the n positions $(q^1, \dots, q^n)$ of our system at some time t together with the n velocities $(\dot{q}_1, \dots, \dot{q}_n)$ at the same time.

Definition 1.42 (Velocity Phase Space). Let Q be the configuration manifold of a physical system. The **velocity phase space** of Q is the tangent bundle TQ .

We also have to specify what we are looking for. In physics, one is interested in an *equation of motion* for the considered system. To this end, a **motion** is a smooth curve

$$\gamma : I \subseteq \mathbb{R} \rightarrow Q.$$

We think about I as a time interval, so γ specifies one evolution of the system Q . The one evolution γ actually taken by the system is what we call our *equation of motion*. To obtain such γ , we have to rely on one empirical axiom.

We do not define what a Lagrangian system is, since this is quite difficult to answer. To our knowledge, it is possible to conjure some Lagrangian yielding equations of motion. However, this is anything but straightforward outside of classical mechanics.

Axiom 1.43 (Hamilton's Principle of Extremal Action). Let Q be a configuration space of a Lagrangian physical system with tangent bundle TQ and Lagrangian $\mathcal{L}$. If and only if a given smooth curve $\gamma : I \rightarrow Q$ is an extremal of the **action functional**

$$S[\mathcal{L}] := \int_I \mathcal{L}(\gamma(t), \dot{\gamma}(t), t) dt,$$

it is a motion of Q .

1.4. LAGRANGIAN MECHANICS

Using this axiom directly to calculate γ from a given Lagrangian is quite cumbersome. The following theorem simplifies this procedure a bit.

Theorem 1.44 (Euler-Lagrange). Let Q be a smooth manifold and $\mathcal{L}$ be a Lagrangian on $TQ \times \mathbb{R}$. A smooth curve $\gamma : I \rightarrow Q$ extremizes the action functional $S[\gamma]$ if and only if the following set of equations is satisfied for all $t \in I$:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\gamma}^i} \right) - \frac{\partial \mathcal{L}}{\partial \gamma^i} = 0 \quad (1.3)$$

where γ_i are the components of γ in some local chart *spanning* TM . These differential equations are called **Lagrange Equations of the second kind**.

Note that while we want to apply this in a physical context, those equations are much more general since we only demand of $\mathcal{L}$ to be smooth. A famous other example of these equations is finding minimal surfaces, found in nature as the thin films soapy water forms on geometrical shapes. To make matters more confusing, one can use these equations to minimize the length functional of the previous section, in which case we call the equations **Euler equations** or, especially if expressed in local coordinates, **Geodesic equations**.

Since our treatment of the novel physical system relies on this theorem, we want to prove it. We are, however, using a lemma without a proof.

Lemma 1.45 (Fundamental Lemma of Variational Calculus). Let $f : (a, b) \rightarrow \mathbb{R}$ be continuous. If

$$\int_a^b f(t) \nu(t) dt = 0$$

is satisfied for all smooth functions $\nu : (a, b) \rightarrow \mathbb{R}$ with compact support, then $f \equiv 0$.

Proof. Since we work with a locally trivializing chart of TQ (meaning the coordinates are a full orthogonal system), the equation $\delta S[\mathcal{L}] = 0$ has to hold for each component γ^i . Hence, all components γ^i have to satisfy this equation separately, reducing the problem to one dimension. Therefore, take a path $\gamma : (t_0, t_1) \rightarrow \mathbb{R}$ (we read γ directly in local coordinates). A variation of γ with fixed endpoints is a smooth function

$$\delta\gamma : (t_0, t_1) \rightarrow \mathbb{R}$$

with $\delta\gamma(t_0) = \delta\gamma(t_1) = 0$. If S has an extremum at γ , the functional derivative vanishes for γ . Using this, we calculate:

$$\begin{aligned} 0 = \delta S[\gamma] &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int_{t_0}^{t_1} \mathcal{L}(\gamma + \epsilon \cdot \delta\gamma, \dot{\gamma} + \epsilon \cdot \dot{\delta\gamma}, t) dt \\ &= \int_{t_0}^{t_1} \left\{ \frac{\partial \mathcal{L}}{\partial \gamma} \cdot \delta\gamma + \frac{\partial \mathcal{L}}{\partial \dot{\gamma}} \cdot \frac{\partial \delta\gamma}{\partial t} + \frac{d\mathcal{L}}{dt} \right\} dt \end{aligned}$$

Concentrating on the second term, we can integrate by parts:

$$\begin{aligned} \int_{t_0}^{t_1} \frac{\partial \mathcal{L}}{\partial \dot{\gamma}} \cdot \frac{\partial \delta\gamma}{\partial t} dt &= \frac{\partial \mathcal{L}}{\partial \dot{\gamma}} \cdot \delta\gamma \Big|_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\gamma}} \cdot \delta\gamma dt \\ &= - \int_{t_0}^{t_1} \frac{\partial \mathcal{L}}{\partial \dot{\gamma}} \cdot \delta\gamma dt. \end{aligned}$$

The first term vanished as $\delta\gamma$ is a variation with fixed endpoints. Plugging this back in yields

$$0 = \int_{t_0}^{t_1} \left\{ \frac{\partial \mathcal{L}}{\partial \gamma} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\gamma}} \right) \right\} \delta\gamma dt.$$

Since $\delta\gamma$ is compactly supported by definition, we apply the fundamental theorem of calculus to obtain our claim.

For the other direction, the proof can practically be read backwards. $\square$

Physical Lagrangians and Constraints

After accepting that nature seems to follow a universal extremal principle, we want to explore the different important forms a Lagrangian can take. The easiest examples are such that $\mathcal{L}$ is connected with potential and kinetic energy.

Unless specified otherwise, we use the Euclidean metric for $\mathbb{R}^{3N}$. This means that every submanifold Q is endowed with the restricted Euclidean metric, hence kinetic energy is well-defined.

Definition 1.46 (Kinetic and Potential Energy). Let TQ be the velocity phase space of a physical system Q . The **kinetic energy** is a quadratic form given locally at $q \in Q$ by

$$T : T_q Q \times \mathbb{R} \rightarrow \mathbb{R}$$

$$\dot{q} \mapsto \frac{\|\dot{q}\|^2}{2}$$

and the **potential energy** is some smooth function

$$U : Q \times \mathbb{R} \rightarrow \mathbb{R}.$$

If the system is autonomous, T and U are independent of the second argument.

Again, there are autonomous and non-autonomous natural systems. All systems we consider will be autonomous, so T and U only depend on t indirectly through the time dependence of a path $(\gamma, \dot{\gamma})$.

If one is not satisfied with this explanation, one can be more physical: Imagine some curve a free particle in a plane can take. Now imagine a constraint as a force acting in one coordinate direction q^1 of the local chart (q^1, q^2) . If we consider this potential to go to infinity, the particle is forced to move in direction q^2 in the limit. It is possible to prove the following proposition by making this idea rigorous.

Definition 1.47 (Natural System). A Lagrangian physical system Q is called **natural** if the Lagrangian is given by

$$\mathcal{L}(q, \dot{q}, t) := T(q, \dot{q}, t) - U(q, t)$$

where $T : TQ \times \mathbb{R} \rightarrow \mathbb{R}$ is a kinetic energy and $U : Q \times \mathbb{R} \rightarrow \mathbb{R}$ is a potential.

Note that in the examples above, many systems are constrained to move on some m -dimensional submanifold Q of $\mathbb{R}^{3N}$. We call such constraints **holonomic** if Q is given by $3N - m$ independent equations of the type

$$f_k(q^1, \dots, q^{3N}, t) = 0$$

with $1 \leq k \leq 3N - m$, $(q^1, \dots, q^{3N})$ being global coordinates of $\mathbb{R}^{3N}$ and f being some smooth function in those coordinates. If f_k is independent of t , we call it **scleronomous**, else it is called **rheonomous**.

Theorem 1.48 (Holonomic Systems). A physical system with purely holonomic constraints is natural.

Example 1.49. A free particle with mass m in $\mathbb{R}^3$ has the kinetic energy

$$T(\dot{q}^1, \dot{q}^2, \dot{q}^3) = \frac{m}{2} \delta^{ij} \dot{q}_i \dot{q}_j.$$

Its Lagrangian is just $\mathcal{L} = T$. The Lagrange equations of the second kind read as

$$0 = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\gamma}^i} \right) = m \ddot{\gamma}^i.$$

Given a set of initial conditions $\gamma^i(t_0) = A_i$ and $\dot{\gamma}^i(t_0) = B_i$, we obtain the motion

$$\gamma^i(t) = A_i t + B_i.$$

Conservation Laws

One huge advantage over the Newtonian formalism is that the Lagrangian directly provides means to connect symmetries of a given system with conservation laws which aid massively in solving for the equations of motion.

Definition 1.50 (Symmetry). Let Q be a configuration manifold with Lagrangian $\mathcal{L}$. We say that Q admits a **symmetry** if there is a one-parameter group of diffeomorphisms

$$\Phi : Q \times \mathbb{R} \times \mathbb{R} \rightarrow M$$

such that

$$\mathcal{L}(d\varphi^s(\mathbf{q}, \dot{\mathbf{q}}, t)) = \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, t)$$

for all $s \in \mathbb{R}$, where we write $\varphi^s(\mathbf{q}, t) := \Phi(\mathbf{q}, t, s)$.

Example 1.51. The most common examples of symmetries are spacial translation

$$\varphi^s(q^1, \dots, q^i, \dots, q^n, t) = (q^1, \dots, q^i + s, \dots, q^n, t)$$

and temporal translation

$$\varphi^s(\mathbf{q}, t) = (\mathbf{q}, t + s).$$

The celebrated theorem of Emmy Noether will connect symmetries to so-called first integrals. If we have a system of ordinary differential equations

$$f(x(t), \dot{x}(t), t) = 0$$

with solutions $x : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^n$, a **first integral** of this equation is a smooth function $\mathcal{I} : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\frac{d}{dt} \mathcal{I}(x(t), t) = 0$$

for any solution x . This means that first integrals are constant along solutions of Lagrangian equations. Their importance arises from the fact that a first integral can always be integrated to reduce the number of unknowns by 1.

Theorem 1.52 (Noether). If the Lagrangian system $(Q, \mathcal{L})$ admits a symmetry Φ , then the Lagrangian equations have a first integral

$$\mathcal{I} : TM \rightarrow \mathbb{R}$$

given in local coordinates by

$$\mathcal{I}(\mathbf{q}, \dot{\mathbf{q}}, t) = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \frac{d\Phi(\mathbf{q})}{ds} \Big|_{s=0}.$$

Example 1.53. We consider the case of translational invariance along some axis. Take N point masses m_i in $\mathbb{R}^3$ with N constraints $f_j(\mathbf{q}_j) = 0$ and a potential $U = U(\mathbf{q})$ which admits translation in q_j^1 -direction. The Lagrangian is given by

$$\mathcal{L} = \sum_i \frac{m_i}{2} \delta^{jk} \dot{q}_{ij} \dot{q}_{ik} + U(\mathbf{q}).$$

The symmetry is given by

$$\varphi^s : q_{i1} \mapsto q_{i1} + s.$$

The first integral is

$$\mathcal{I} = \sum_i m_i \dot{q}_{i1} = \sum_i p_{i1}$$

where p_{i1} is the mechanical momentum of the i -th particle in direction of q^1 . This shows that translational symmetry corresponds to conservation of linear, mechanical momentum.

1.5 Non-holonomic Systems and d'Alemberts Principle

Since the system we are considering is a complication of a very well-known, classical non-holonomic system, we have to consider how to deal with such systems. It turns out that while non-holonomic constraints can be formulated in an adapted version of Lagrange's formalism, d'Alemberts principle, which is equivalent to Lagrange's principle, has certain advantages depending on the particular problem at hand.

1.5.1 Euler-Lagrangian Mechanics

Looking back at our system with $3N - m$ degrees of freedom, we use the m holonomic constraints $f_k(\mathbf{q}, t) = 0$ to eliminate m coordinates by integration. This yields the orthogonal trivialization of TQ which we needed to justify that the coordinate components in the Euler-Lagrange equation vanish independently of each other. Now, we are confronted with m non-holonomic constraints

$$g_k(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \dots, t) = 0$$

which employ a dependence of the coordinates on one another without being integrable. We will not be able to solve this problem in general, but there are noteworthy special cases.

If we assume that the constraints are still holonomic but we did not integrate

1.5. NON-HOLONOMIC SYSTEMS AND D'ALEMBERTS PRINCIPLE

them and therefore still have $\dim Q$ coordinates and m first integrals, we can use Lagrange multipliers to save our principle of extremal action.

Theorem 1.54 (Lagrangian Multipliers). Let $(Q, \mathcal{L})$ be a Lagrangian system with m holonomic constraints $f(\mathbf{q}, t) = 0$. In this case, a smooth path $\gamma : I \rightarrow Q$ is an extremal of S if and only if the **Lagrange Equations of the First Kind**

$$\frac{\partial \mathcal{L}}{\partial \gamma^i} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\gamma}^i} \right) + \sum_{k=1}^m \lambda_k(t) \frac{\partial f_k}{\partial \gamma^i} = 0$$

are satisfied, where the $\lambda_k(t)$ are Lagrangian multipliers. The term

$$Q_i = \sum_{k=1}^m \lambda_k(t) \frac{\partial f_k}{\partial \gamma^i}$$

is called **force of constraint** or **generalized force**.

This notion of a force of constraint fits nicely with our earlier intuitive explanation that constraints can be considered as very strong force fields.

We can apply this equation for a very specific set of non-holonomic constraints.

Definition 1.55 (Semi-Holonomic Constraint). Let $(Q, \mathcal{L})$ be a Lagrangian system with m constraints of the form

$$\omega_k = a_{ik}(\mathbf{q}, \dot{\mathbf{q}}, t) dq^i + a_{k0}(\mathbf{q}, \dot{\mathbf{q}}, t) dt = 0$$

where $1 \leq k \leq m$ and the ω_k are Pfaffian forms. In this case, the constraints are called **semi-holonomic**.

Unfortunately, this is not enough to apply the Lagrange equations of the first kind.

Theorem 1.56 (Integrable Semi-Holonomic Constraints). A Lagrangian system $(Q, \mathcal{L})$ with a semi-holonomic constraint ω is equivalent to a holonomic system if and only if there is an integrating factor $\lambda \neq 0$ and a one-form η such that

$$\lambda \cdot \omega + f \cdot \eta = df,$$

where f is the holonomic constraint and df is the exterior derivative of f .

Example 1.57. A simple example is a hoop rolling down a two-dimensional incline of height h and angle α without slipping. The obvious choice of generalized coordinates are the distance from the starting point and the rotational angle of the hoop, so our first guess at the configuration manifold is $\mathbb{R} \times \mathbb{S}^1$. The kinetic energy is the sum of rotational and translational energy

$$T = \frac{m}{2} (\|\dot{\mathbf{q}}\|^2 + R^2 \|\dot{\theta}\|^2)$$

and the potential is dependent on the height, hence also on q :

$$U = (h - q \sin \alpha)mg.$$

The rolling condition reads as

$$\dot{q} - R\dot{\theta} = 0$$

which is equivalent to the Pfaffian form

$$0 = dq - R d\theta$$

and hence semi-holonomic. One can immediately see that this is integrable, but for the sake of demonstration, we use one Lagrangian multiplier. The modified Lagrangian is

$$\mathcal{L}(q, \theta, t, \lambda) = \frac{m}{2}(\|\dot{q}\|^2 + R^2\|\dot{\theta}\|^2) + (q \sin \alpha - h)mg + \lambda(\dot{q} - R\dot{\theta})$$

and the Lagrange equations of the second kind yield

$$m\ddot{q} - mg \sin \alpha + \lambda = 0$$

and

$$mR^2\ddot{\theta} - \lambda R = 0.$$

Together with our equation of constraint $R\ddot{\theta} = \ddot{q}$, we obtain the following family of solutions

$$\begin{cases} \ddot{q} = \frac{g \sin \alpha}{2} \\ \ddot{\theta} = \frac{g \sin \alpha}{2R} \\ \lambda = \frac{mg \sin \alpha}{2} \end{cases}.$$

1.5.2 Lagrange-d'Alembertian Mechanics

We turn our attention to a slightly different approach closer to Newton's original formulation, but nonetheless more general. Consider a system with n -dim. configuration space subject to $m < n$ constraints of the form

$$\sum_{k=1}^n a_k^j(q^i) \dot{q}^k = A(q) \dot{q} = 0.$$

In this particular formulation, the constraints are holonomic if we find a constraint vector $\{b^j(q^i)\}$ with $1 \leq j \leq m$ such that

$$\sum_{k=1}^n (\partial_k b^j) \dot{q}^k = 0,$$

i.e. $\partial_k b^j = a_k^j$. We are mostly interested in the case where this is not possible, the non-holonomic case.

We call the $m \times n$ -matrix $A(q) = \{a_k^j(q^i)\}$ with $1 \leq k \leq n$ and $1 \leq j \leq m$ the constraint matrix and demand that it is at most linearly dependent on the velocities $\dot{q}^i$. We imagine that the system obeys such constraints only because there are forces acting upon it; denote these constraint forces by $F = (F_1, \dots, F_n)$. Now we imagine a very small displacement of the system in the configuration space, called **virtual displacement** and denoted $\delta q = (\delta q^1, \dots, \delta q^n)$.

Definition 1.58 (Constrained Space). Given a Lagrangian system $(Q, (q^i)_{1 \leq i \leq n})$ with velocity phase space TQ and constraints $A(q)\dot{q} = 0$ as above, a virtual displacement δq is a **constrained displacement** if

$$\delta q \in \mathcal{D}_q := \{\delta q \in T_q Q \mid A(q)\delta q = 0\}.$$

We call $\mathcal{D}_q$ the constrained space.

1.5. NON-HOLONOMIC SYSTEMS AND D'ALEMBERTS PRINCIPLE

Along such a displacement, the work needed is called virtual work and defined as

$$\delta W := \langle F, \delta q \rangle = F_i \delta q^i.$$

To derive the equations of motion, we need the axiom of this formulation:

Axiom 1.59 (d'Alemberts Principle). Along any virtual constrained displacement, virtual work vanishes:

$$\delta W = 0.$$

Note that if the δq^i and the row vectors of A are linearly independent, F has to be a linear combination of the rows of A , so

$$F = \Lambda A(q).$$

The row-vector $\Lambda = (\lambda_1, \dots, \lambda_m)$ consists of Lagrangian multipliers λ_i (which are, confusingly, similar in function, but technically not the same Lagrangian Multipliers as the ones in optimization problems from before). So far, external forces which are not constraints are still missing, so denote these as $F^e = (F_1^e, \dots, F_n^e)$. With this, we are able to formulate our theorem for the equations of motion:

Theorem 1.60 (Lagrange-d'Alembert Principle). Let $(Q, (q^i))$ be a Lagrangian system with Lagrangian $\mathcal{L} = T - U$, where T is the kinetic energy and U are potentials. If the system is subject to constraints $A(q)\dot{q} = 0$ as defined above, and to external forces F^e , any virtual displacement along a motion $t \mapsto q(t)$ extremizes the Lagrange-d'Alembert action

$$\delta \int_I \mathcal{L}(q(t), \dot{q}(t)) dt + \int_I \langle F^e, \delta q \rangle dt = 0 \quad (1.4)$$

In this case, this theorem is actually not the most natural way of arriving at the equations of motions. Assuming just d'Alembert's Principle and Newton's axioms, one derives:

Theorem 1.61 (Lagrange-d'Alembert Equations of Motion). Let $(Q, (q^i))$ be a Lagrangian system with Lagrangian $\mathcal{L} = T - U$, where T is the kinetic energy and U are potentials. If the system is subject to constraints $A(q)\dot{q} = 0$ as defined above, and to external forces F^e , any physical motion $t \mapsto q(t)$ satisfies the Lagrange-d'Alembert Equations of motion:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^i} - \frac{\partial \mathcal{L}}{\partial q^i} = \lambda_j a_i^j + F_i^e. \quad (1.5)$$

One can show that these equations are precisely equivalent to the variational formulation for the action integral. The main difference to the Euler-Lagrange approach is the order in which we impose the constraints: In this case, we impose the constraints before doing any variation, so we constrain the admissible variations themselves. In the Euler-Lagrange case, we impose the constraints by using a modified Lagrangian incorporating the Lagrangian

Why should this axiom be true? Intuitively, the same idea as before applies: The constraint forces are imagined to be very strong, so they practically force the particle to move orthogonally to the force. A nice mental image would be an MRI magnet: If you tried to move a little ferromagnetic ball in such a strong field, every movement increasing the magnetic energy will be almost impossible. However, one needs to remember that q^i may not only consist of path-like components, but for example also angles or even temperature. Since work is calculated as integral over the scalar product of the forces and the path taken, it is expected to vanish if we only move orthogonal to constraint forces.

Note that one has not only these n equations, but also m equations of the form $0 = A(q)\dot{q}(t)$ which can sometimes be used to eliminate the Lagrangian multipliers.

Multipliers, so we actually constrain the velocities of admissible variations. These two ways of constraining do sometimes commute, but not in general. As it turns out, the Lagrangian-d'Alembert equations predict the correct dynamical mechanical behaviour, while the variational problem answers the question of optimal control.

1.6 Ordinary Differential Equations

This section is just meant to quickly list the differential equations we will encounter later and give methods to solve them. The general solvability is governed by the following property and theorem.

Definition 1.62 (Lipschitz Continuity). Let $D \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be an open subset and $f : D \rightarrow \mathbb{R}^k$ be a function. If there exists a constant $L \geq 0$ such that

$$\|f(s, x) - f(s, y)\| \leq L\|x - y\|$$

for all $(s, x), (s, y) \in D$, we call f **globally Lipschitz continuous** with respect to x with **Lipschitz constant** L . If for any $p \in D$ there is a neighbourhood U of p such that f restricted to $U \cap D$ is Lipschitz, we call f **locally Lipschitz (continuous)**.

Theorem 1.63 (Picard-Lindelöf). Let $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be an open subset and $f : D \rightarrow \mathbb{R}^n$ be a continuous function which is Lipschitz continuous w.r.t. x . Then any initial value problem of the form

$$\left\{ \begin{array}{l} \dot{x} = f(t, x) \\ x(t_0) = x_0 \\ (t_0, x_0) \in D \end{array} \right\}$$

is locally uniquely solvable.

In this, and only in this, chapter we will do separation of variables in ODEs by seemingly separating differentials of the form $\frac{dx}{dt}$.

Autonomous ODE

Definition 1.64 (Second-Order Autonomous ODE). A **second-order autonomous first-order ODE** is of the form

$$\ddot{x}(t) = f(x, \dot{x}) \tag{1.6}$$

where $x : \Omega \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is at least twice differentiable and $f : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ is continuous.

We consider the special case where $f(x, \dot{x}) = f(x)$. The substitution $y := \frac{dx}{dt}$ yields

$$\int y \, dy = \int f(x) \, dx \implies v = \pm \sqrt{2 \int f(x) \, dx + C_1}$$

1.6. ORDINARY DIFFERENTIAL EQUATIONS

where $C_1 \in \mathbb{R}$ is some constant of integration. Plugging back yields the implicit solution set

$$\pm \int \frac{dx}{\sqrt{2 \int f(x) dx + C_1}} = t + C_2 \quad (1.7)$$

with a second constant $C_2 \in \mathbb{R}$.

CHAPTER TWO

Magnetic Ball on an Incline

2.1 Two-Dimensional System

Lagrangian Equation of Motion

We start our investigation by considering a very simplified model of a rigid two-dimensional disk of mass M with a constant magnetic moment $\mathbf{m}$ rolling on an inclined plane of length l and angle α without slipping in the earth magnetic field $\mathbf{B}$. The moment of inertia is $I = \frac{MR^2}{2}$. Similar to the example of semi-holonomic constraints, our generalized coordinates are the rolling distance q and the rotational angle θ . Note that while $\mathbf{m}$ is constant in the rolling frame of the disk, the angle between $\mathbf{m}$ and $\mathbf{B}$ changes with the rolling motion. We have again the same semi-holonomic constraint

$$dq - R d\theta = 0.$$

This is a first integral, yielding

$$q(t) - q_0 = R\theta(t) - \theta_0$$

where we set $q_0 := q(0)$ and $\theta_0 := \theta(0)$. For simplicity, we choose the coordinates in such a way that w.l.o.g. $q_0 = 0 = \theta_0$. The kinetic energy is given by

$$T(q, \dot{q}) = \frac{M}{2} \|\dot{q}\|^2 + \frac{I}{2} \|\dot{\theta}\|^2 = \frac{3M}{4} \dot{q}^2$$

where we substituted $\theta = \frac{q}{R}$. Two potentials act on the disk, the first being gravity

$$U_g(q) = Mgh = (q - l)Mg \sin \alpha.$$

The second is due to the magnetic moment of the disk interacting with the earth's magnetic field. Assuming an ideal magnetic dipole interaction, the potential is governed by the angle between $\mathbf{m}$ and $\mathbf{B}$. Assuming an initial angle of β at $t = 0$, the angle at time t is simply $\beta - \theta$. Therefore, the electromagnetic potential is

$$U_{em}(q) = -\langle \mathbf{m}, \mathbf{B} \rangle = -mB \cos \left(\beta - \frac{q}{R} \right)$$

with $m = \|\mathbf{m}\|$ and $B = \|\mathbf{B}\|$.

We obtain the Lagrangian

$$\mathcal{L}(q, \dot{q}) = T - U_g - U_{em} = \frac{3M}{4} \|\dot{q}(t)\|^2 + (l - q(t))Mg \sin \alpha + mB \cos \left(\beta - \frac{q(t)}{R} \right)$$

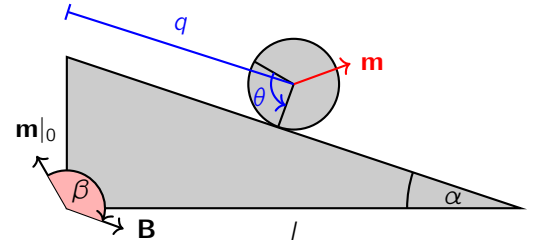


Figure 2.1: A sketch of the considered system. The vector $\mathbf{m}|_0$ denotes the magnetic moment at $t = 0$. One can see that the angle between $\mathbf{m}$ and $\mathbf{B}$ at any point in time is the difference between β and θ . Since $\cos$ is symmetric on $\mathbb{R}$, the change of sign as $\mathbf{m}$ passes $\mathbf{B}$ is not of concern.

Allowing β to be arbitrary is probably not necessary for the practical experiment as the dipole will align itself with the outer field if one does not impair this movement until the disk starts rolling.

which gives rise to just one Lagrangian equation of the second kind:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0 \quad (2.1)$$

This yields the ODE

$$\ddot{q}(t) = \frac{2g}{3} \sin \alpha + \frac{2mB}{3MR} \sin \left(\beta - \frac{q}{R} \right) = f(q(t))$$

which is a second-order autonomous ODE. We already discussed how to solve such an equation. The first integral needed is

$$\begin{aligned} \int_{q_0=0}^q f(\tau) d\tau &= \int_{q_0=0}^q \left(\frac{2g}{3} \sin \alpha + \frac{2mB}{3MR} \sin \left(\beta - \frac{q}{R} \right) \right) d\tau \\ &= \frac{2g}{3} \sin(\alpha) q - \frac{2mB}{3M} \left(\cos \left(\beta - \frac{q}{R} \right) - \cos \beta \right) \end{aligned}$$

We make some abbreviations to keep everything readable. Define $A := \frac{4g}{3} \sin \alpha$, $B := -\frac{4mB}{3M}$ and $C := \frac{4mB}{3M} \cos \beta$. The implicit solution is then given by

$$t = \pm \int_0^{q(t)} \frac{1}{\sqrt{Aq - B \cos(\beta - \frac{q}{R}) + C}} dq. \quad (2.2)$$

Unfortunately, this integral has no closed-form solution. The only option is to calculate the integral numerically, for example by means of a Taylor series, and then invert pointwise to obtain $q = q(t)$. We will focus on some aspects which can be derived in limiting cases.

Symmetries

The derived Lagrangian is not very abundant in symmetries. As $\mathcal{L}$ is autonomous, the system admits time translation symmetry and hence conservation of energy. However, this is not very surprising as we did not enforce the rolling condition by means of a dissipative Lagrangian but instead as semi-holonomic integrable constraint. The variable q does appear explicitly in the Lagrangian, so it can not be cyclic. The only non-trivial symmetry remains time symmetry.

Conclusion

While it is quite surprising that such a simple system does not admit a closed-form analytic solution, we gain some first insight in the general usefulness of the Lagrangian formalism to derive equations of motion. However, our hope to solve the more complicated three-dimensional system is very much reduced, especially since the three-dimensional rolling constraint is not always semi-holonomic.

2.2 Three-dimensional System

2.2.1 Euler-Lagrangian Mechanics

We want to address the more complicated problem of a three-dimensional, spherical, solid magnetic dipole moving on an inclined plane in earth's magnetic

2.2. THREE-DIMENSIONAL SYSTEM

field. Following [rundquist], we are dropping the restriction of the plane being an incline and consider two-dimensional smooth submanifolds of $\mathbb{R}^2$ as subsurface for the rolling ball. The system is characterized fully by three coordinates $(q^1, q^2, q^3) \in \mathbb{R}^3$ describing the center-of-mass motion with respect to a fixed coordinate frame and three Euler angles $(\phi, \psi, \theta) \in T^3$ describing the rotation of the body coordinate system. Hence, the configuration space is given by

$$Q = \mathbb{R}^3 \times \text{SO}(3).$$

Taking a path $t \mapsto (q^1(t), q^2(t), q^3(t), \phi(t), \psi(t), \theta(t))$, we obtain the linear kinetic energy

$$T_{\text{lin}}(\dot{q}^1, \dot{q}^2, \dot{q}^3) = \frac{M}{2} \|\dot{\mathbf{q}}\|^2$$

and the rotational kinetic energy

$$T_{\text{rot}}(\phi, \psi, \theta, \dot{\phi}, \dot{\psi}, \dot{\theta}) = \frac{I}{2} \|\boldsymbol{\omega}\|^2$$

where we wrote $\mathbf{q} := (q^1, q^2, q^3)$ and $\boldsymbol{\omega} := (\omega_1, \omega_2, \omega_3)$ for brevity. Assume that the hypersurface restricting the motion of the ball in q^3 -direction is given by an implicit equation

$$f(q^1(t), q^2(t)) = q^3(t).$$

Then, the gravitational potential is

$$U_{\text{grav}}(q^1, q^2) = M \cdot g \cdot f(q^1(t), q^2(t)).$$

The magnetic moment of the ball is a vector $\mathbf{m} \in \mathbb{R}^3$ with $m := \|\mathbf{m}\|$ fixed in the rotating body frame but rotating from the view of the stationary frame. The magnetic field of the earth is a constant vector $\mathbf{B} \in \mathbb{R}^3$ with $B := \|\mathbf{B}\|$. Assuming ideal magnetic dipole interaction, we need the scalar product $\langle \mathbf{m}, \mathbf{B} \rangle$ to get an expression for U_{em} in Euler angles. For this, choose $(\phi_0, \psi_0, \theta_0)$ such that the body z -axis is aligned with the magnetic moment: $\mathbf{z} \perp \mathbf{m}$. Using that Euler angles are a local parametrization of $\text{SO}(3)$, we can obtain the direction of the rotating z -axis at time t by the action of the corresponding rotation matrix on $\mathbf{z}|_{t=0}$. The scalar product then decomposes into the easier form $mB \langle A_t(\phi, \psi, \theta) \mathbf{z}_{t=0}, \mathbf{B} \rangle$ with A_t being a one-parameter subgroup corresponding to the chosen path, and we obtain

$$\begin{aligned} U_{\text{em}}(\phi, \psi, \theta) &= -\langle \mathbf{m}, \mathbf{B} \rangle = -mB \langle A_t \mathbf{z}_{t=0}, \mathbf{B} \rangle \\ &= -m(c_\psi s_\theta B_1 + s_\psi s_\theta B_2 + c_\theta B_3) = U_{\text{em}}(\psi, \theta). \end{aligned}$$

The rolling constraint yields

$$\|\boldsymbol{\omega}\| = \frac{\|\dot{\mathbf{q}}\|}{R}$$

and hence we can simplify

$$T_{\text{rot}}(\mathbf{q}) = \frac{I}{2R^2} \|\dot{\mathbf{q}}\|,$$

In our case, we are interested in a very simple surface describing a ramp in the (q^1, q^2) -plane. The most naive way to describe this is to restrict the configuration space to be the right half-space

$$\mathbb{R}_+^3 = \{(x^1, x^2, x^3) \in \mathbb{R}^3 \mid x^2 \geq 0\}.$$

Then, we can define a continuous function $f : \mathbb{R}_+^3 \rightarrow \mathbb{R}$ given by

$$f(q^2) := \begin{cases} \alpha q^2 + \beta & q^2 > \frac{\beta}{\alpha} \\ 0 & q^2 \leq \frac{\beta}{\alpha} \end{cases}$$

where $\alpha \in \mathbb{R}$ is the slope and $\beta \in \mathbb{R}$ is the maximal height of the ramp. Clearly, this f cannot be smooth at $q^2 = \frac{\beta}{\alpha}$ which will be a problem if we have to take directional derivatives later. While an ideal ramp will always suffer from a non-differentiable line parallel to the q^1 -axis, we can replace it by the smooth function

$$f_\delta(q^2) := \begin{cases} -\frac{\alpha x - \beta}{1 - \exp(-\delta(\alpha x - \beta))} & q^2 \neq \frac{\beta}{\alpha} \\ \frac{1}{\alpha} & q^2 = \frac{\beta}{\alpha} \end{cases},$$

where for $\delta \rightarrow \infty$, we recover the ideal ramp with slope α and q^2 -intersection β .

Note that while we can eliminate $\boldsymbol{\omega}$ completely, we may need to recover this vector if we aspire to animate the results. While we could just solve for the Euler angles, it is easier to note that the rolling condition also guarantees that

$$\dot{\mathbf{q}} \perp \boldsymbol{\omega} \perp \mathbf{n},$$

where $\mathbf{n}$ is the surface normal. Defining an auxilliary function

$$F(\mathbf{q}) := q^3 - f(q^1, q^2)$$

which is the implicit equation of the surface, we can obtain the normal as

$$\mathbf{n} = \frac{\text{grad } F}{\|\text{grad } F\|}.$$

With this, $\boldsymbol{\omega}$ can be fully reconstructed from the cartesian part of the configuration space:

$$\boldsymbol{\omega} = \frac{\|\dot{\mathbf{q}}\|}{R} \left(\frac{\text{grad } F}{\|\text{grad } F\|} \times \dot{\mathbf{q}} \right).$$

eliminating the dependence on the Euler angles. Using $I = \frac{2MR^2}{5}$ for an ideal ball, we can denote the Lagrangian as

$$\begin{aligned} \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, \psi, \theta) &= U_{\text{grav}} + U_{\text{em}} - T_{\text{kin}} - T_{\text{rot}} \\ &= Mgf(q^1, q^2) + -m(c_\psi s_\theta B_1 + s_\psi s_\theta B_2 + c_\theta B_3) - \frac{7M}{10} \|\dot{\mathbf{q}}\|^2 \end{aligned} \quad (2.3)$$

This yields five Lagrangian equations of the second kind:

$$0 = \frac{7M}{5} \dot{q}^1 + M g \partial_1 f = \frac{7M}{5} \dot{q}^1 \quad (2.4)$$

$$0 = \frac{7M}{5} \dot{q}^2 + M g \partial_2 f \approx \frac{7M}{5} \dot{q}^2 + \alpha \quad (2.5)$$

$$0 = \frac{7M}{5} \dot{q}^3 \quad (2.6)$$

$$0 = m(-s_\psi s_\theta B_1 + c_\psi s_\theta B_2) \quad (2.7)$$

$$0 = m(c_\psi s_\theta B_1 + s_\psi c_\theta B_2 - s_\theta B_3) \quad (2.8)$$

where we took the limit $\delta \rightarrow \infty$ in f away from the non-differentiable points. Unfortunately, the high non-linear nature of the equations makes simulations very difficult, so we considered another approach.

2.2.2 Non-Holonomic Mechanics: The Rigid Ball

We follow [.] Consider a perfect homogeneous solid ball of mass M , radius R , and inertia scalar $I = \frac{2}{5}MR^2$ rolling without slipping on an incline of slope $\alpha \in \mathbb{R}$. Write $\mathbf{r} = (x, y, z)^\top \in \mathbb{R}^3$ for the bodies inertial coordinates and $P \in \text{SO}(3)$ for the rotational coordinates, $\boldsymbol{\Omega} = (\Omega_1, \Omega_2, \Omega_3)^\top$ for the bodies inertial rotational velocity, and $\boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)^\top$ for the rotational velocity in the body frame. Furthermore, given a vector $\mathbf{v} = (v_1, v_2, v_3)^\top \in \mathbb{R}^3$, we define the associated cross-product operator

$$\hat{\mathbf{v}} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

as matrix operator

$$\hat{\mathbf{v}} := \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix}.$$

Note that this matrix is skew-symmetric. Strictly speaking, this operator arises by the isomorphism $\mathfrak{so}(3) \cong \mathbb{R}^3$.

Denote by $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ the canonical Euclidean basis of $\mathbb{R}^3$. Our goal is to derive analytical equations of motion for $\mathbf{r}(t)$ and $P(t)$ under the influence of gravity $U = Mgz$. The incline is described as a zero set of $F(y, z) = z - \alpha y$, yielding the surface normal field

$$\mathbf{n} = \frac{\text{grad } F}{\|\text{grad } F\|} = \frac{(0, -\alpha, 1)}{\sqrt{\alpha^2 + 1}}.$$

The Lagrangian of this system is evidently given as

$$\mathcal{L} = T - U = \frac{M}{2} \|\dot{\mathbf{r}}\|^2 + \frac{I}{2} \|\boldsymbol{\Omega}\|^2 - Mgz. \quad (2.9)$$

The virtual displacements are $\delta \mathbf{r}$ and δP . Since a virtual rotation is induced by the cross product on $\mathfrak{so}(3)$, we can introduce the virtual rotation vector

2.2. THREE-DIMENSIONAL SYSTEM

$\delta\theta$ such that $\delta P = \widehat{\delta\theta}(P)$. It is very useful to split the rotational degree of freedoms into parts normal and tangential to the incline. Write

$$\Omega = \Omega_{\perp} \mathbf{n} + \Omega_{\parallel}$$

with $\Omega_{\perp} = \langle \Omega, \mathbf{n} \rangle$ and $\Omega_{\parallel} = \Omega - \Omega_{\perp} \mathbf{n}$. The same splitting also applies to the angular variation

$$\delta\theta = \delta\theta_{\perp} \mathbf{n} + \delta\theta_{\parallel}.$$

Since the ball rolls without slipping, we get the constraint

$$\dot{\mathbf{r}} = R\Omega \times \mathbf{n}.$$

This implies that the admissible variations are restricted by

$$\delta\mathbf{r} = R\delta\theta \times \mathbf{n} = R(\delta\theta_{\perp} \mathbf{n} + \delta\theta_{\parallel}) \times \mathbf{n} = R\delta\theta_{\parallel} \times \mathbf{n} = R\widehat{\delta\theta_{\parallel}}(\mathbf{n}).$$

With this, we can either do a variation of $\mathcal{L}$ or use the d'Alembertian principle directly since the forces are easily derived as

$$\mathbf{F}_r = -M\dot{\mathbf{r}} - Mg\mathbf{e}_3$$

and

$$\mathbf{F}_{\theta} = -I\dot{\Omega}.$$

Now we calculate virtual work under the assumption of d'Alembert's principle:

$$\begin{aligned} 0 = \delta W &= \langle \mathbf{F}_r, \delta\mathbf{r} \rangle + \langle \mathbf{F}_{\theta}, \delta\theta \rangle = \langle -M(\dot{\mathbf{r}} + g\mathbf{e}_3), \delta\mathbf{r} \rangle - \langle I\dot{\Omega}, \delta\theta \rangle \\ &= \langle -M(\dot{\mathbf{r}} + g\mathbf{e}_3), R\delta\theta \times \mathbf{n} \rangle - \langle I\dot{\Omega}, \delta\theta \rangle \\ &= \langle MR\mathbf{n} \times (-\dot{\mathbf{r}} - g\mathbf{e}_3) - I\dot{\Omega}, \delta\theta \rangle \end{aligned}$$

We used $\langle \mathbf{a}, \mathbf{b} \times \mathbf{c} \rangle = \langle \mathbf{b}, \mathbf{c} \times \mathbf{a} \rangle$ and symmetry of the scalar product.

We know that the scalar product is non-degenerate, so for arbitrary $\delta\theta$, it can only vanish if the first argument vanishes. This yields

$$I\dot{\Omega} = MR\mathbf{n} \times (-\dot{\mathbf{r}} - g\mathbf{e}_3).$$

Differentiating the rolling constraint, we obtain

$$\dot{\mathbf{r}} = R\dot{\Omega} \times \mathbf{n} + R\Omega \times \dot{\mathbf{n}} = R\dot{\Omega} \times \mathbf{n}.$$

Plugging in yields

$$I\dot{\Omega} = RM\mathbf{n} \times (R\dot{\Omega} \times \mathbf{n} + g\mathbf{e}_3) = -R^2 M\dot{\Omega}_{\parallel} - RMg\mathbf{n} \times \mathbf{e}_3.$$

We used

We define projection operators $\pi_{\perp} = \mathbf{nn}^T$ onto $\text{span}\{\mathbf{n}\}$ and $\pi_{\parallel} = \text{id} - \mathbf{nn}^T$ onto $\text{span}\{\mathbf{n}\}^{\perp}$. The tangential equation reads

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = -\langle \mathbf{c}, \mathbf{b} \rangle \mathbf{a} + \langle \mathbf{c}, \mathbf{a} \rangle \mathbf{b}$$

and $\dot{\Omega}_{\perp} \mathbf{n} - \dot{\Omega} = \dot{\Omega}_{\parallel}$.

$$I\pi_{\parallel}\dot{\Omega} = I\dot{\Omega}_{\parallel} = -R^2 M\dot{\Omega}_{\parallel} - RMg\mathbf{e}_3 \times \mathbf{n}.$$

Rearranging yields

$$\dot{\Omega}_{\parallel} = \frac{RMg}{I + R^2 M} \mathbf{n} \times \mathbf{e}_3.$$

This means that angular velocity parallel to the incline changes with a constant factor determined by the incline and the sphere's parameters. The normal equation is

$$\pi_{\perp}\dot{\Omega} = \dot{\Omega}_{\perp} = 0,$$

so the angular velocity normal to the incline is conserved. Substituting back into the rolling constraint, we obtain

$$\dot{\mathbf{r}} = R\dot{\boldsymbol{\Omega}} \times \mathbf{n} = \frac{RMg}{I + R^2M}(\mathbf{e}_3 - \langle \mathbf{n}, \mathbf{e}_3 \rangle \mathbf{n}).$$

Hence the linear acceleration is constant, the sphere moves like a mass point at the center of gravity. Collecting everything, we have the complete solution

$$\left\{ \begin{array}{l} \dot{\mathbf{r}} = \frac{R^2Mg}{I + R^2M}(\mathbf{e}_3 - \langle \mathbf{n}, \mathbf{e}_3 \rangle \mathbf{n}) \\ \dot{P}(t) = \hat{\boldsymbol{\Omega}}(P(t)) \\ \dot{\boldsymbol{\Omega}}_{\parallel} = \frac{RMg}{I + R^2M}(\mathbf{n} \times \mathbf{e}_3) \\ \dot{\boldsymbol{\Omega}}_{\perp} = 0 \\ \boldsymbol{\omega} = P(t)^{\top} \boldsymbol{\Omega} \end{array} \right\}. \quad (2.10)$$

For the case at hand, we can restrict $\alpha \in [0, \frac{\pi}{2})$ and substitute $\alpha = \tan(\varphi)$, so $\mathbf{n} = \frac{(0, -\tan(\varphi), 1)^{\top}}{\sec(\varphi)} = (0, -\sin(\varphi), \cos(\varphi))^{\top}$, yielding

$$\left\{ \begin{array}{l} \dot{\mathbf{r}} = \frac{5g}{7}(0, \sin(\varphi) \cos(\varphi), 1 - \cos(\varphi))^{\top} \\ \dot{\boldsymbol{\Omega}}_{\parallel} = -\frac{5g \sin(\varphi)}{7} \mathbf{e}_1 \\ \dot{\boldsymbol{\Omega}}_{\perp} = 0 \\ \boldsymbol{\omega} = P^{\top}(t) \boldsymbol{\Omega} \\ \dot{P}(t) = \hat{\boldsymbol{\Omega}}(P(t)) \end{array} \right\}.$$

We see that $\mathbf{r}(t)$ is determined by a second-order ODE and $\boldsymbol{\Omega}$ is determined by a first-order ODE, so we are free to choose three vector-valued initial conditions $\mathbf{r}_0, \mathbf{v}_0, \boldsymbol{\Omega}_0 \in \mathbb{R}^3$. The initial value problem is then uniquely solvable by Picard-Lindelöf's Theorem with solution

$$\left\{ \begin{array}{l} x(t) = v_{x,0}t + x_0 \\ y(t) = -\frac{5g}{14} \sin(\varphi) \cos(\varphi) t^2 + v_{y,0}t + y_0 \\ z(t) = \frac{5g}{14} (1 - \cos(\varphi)) t^2 + v_{z,0}t + z_0 \\ \Omega_1(t) = -\frac{5g}{7} \sin(\varphi) t + \Omega_{1,0} \\ \Omega_2(t) = \Omega_{2,0} \\ \Omega_3(t) = \Omega_{3,0} =: s \end{array} \right\}. \quad (2.11)$$

In the case where all initial values are chosen to vanish, we see that motion is governed by gravitational force in the same way as it is for a point mass. Rotation is only caused by gravitational acceleration in the corresponding direction, while all other rotations are conserved.

Versor Representation For computational reasons, we want to convert rotational matrices to versors. Instead of a rotational matrix $P \in \text{SO}(3)$, we describe the orientation now by a versor $\mathbf{q} \in \mathbb{S}^3$ and identify all vectors $\mathbf{v} \in \mathbb{R}^3$ with $(0, \mathbf{v}) \in \mathfrak{S}\mathbb{H} \subset \mathbb{H}$, their corresponding pure quaternion. We obtain:

$$\left\{ \begin{array}{l} (0, \dot{\mathbf{r}}) = \frac{R^2Mg}{I + R^2M}((0, \mathbf{e}_3) + \Re((0, \mathbf{n})(0, \mathbf{e}_3))(0, \mathbf{n})) \\ \dot{\mathbf{q}}(t) = \frac{1}{2}(0, \boldsymbol{\Omega})\mathbf{q} \\ \dot{\boldsymbol{\Omega}}_{\parallel} = \frac{RMg}{I + R^2M}(\Im((0, \mathbf{n})(0, \mathbf{e}_3))) \\ \dot{\boldsymbol{\Omega}}_{\perp} = 0 \\ \boldsymbol{\omega} = \Im(\mathbf{q}^{-1}(0, \boldsymbol{\Omega})\mathbf{q}) \end{array} \right\}.$$

Note that we made heavy use of $(0, \mathbf{v})(0, \mathbf{w}) = (-\langle \mathbf{v}, \mathbf{w} \rangle, \mathbf{v} \times \mathbf{w})$.

2.2.3 Non-Holonomic Mechanics: The Magnetic Sphere

Keeping all notations and parameters from the previous subsection, we add a permanent magnetic dipole $\boldsymbol{\mu} \in \mathbb{R}^3$ to our solid sphere and introduce a constant homogeneous magnetic field $\boldsymbol{B} \in \mathbb{R}^3$. Note that there is now a subtle difference in the two frames: In the body frame, $\boldsymbol{\mu}$ is a constant vector revolving with the sphere, but the magnetic field moves from the perspective of the ball, so $\boldsymbol{B}_{\text{body}}(t) = P(t)^\top \boldsymbol{B}$. Viewing the situation from the inertial frame, $\boldsymbol{B}$ is constant, but $\boldsymbol{\mu}_{\text{inert}}(t) = P(t)\boldsymbol{\mu}$. We obtain a potential $U_{\text{em}}(t) = -\langle \boldsymbol{\mu}, \boldsymbol{B} \rangle$ and a new Lagrangian

$$\mathcal{L} = T - U = \frac{M}{2} \|\dot{\boldsymbol{r}}\|^2 + \frac{I}{2} \|\boldsymbol{\Omega}\|^2 - Mgz + \langle \boldsymbol{\mu}, \boldsymbol{B} \rangle. \quad (2.12)$$

Again, we can skip the variational step and use Newton's law to obtain the new forces

$$\boldsymbol{F}_r = -M\ddot{\boldsymbol{r}} - Mg\boldsymbol{e}_3$$

and

$$\boldsymbol{F}_\theta = -I\dot{\boldsymbol{\Omega}} + \boldsymbol{\mu} \times \boldsymbol{B}.$$

Our previous calculation still applies, so d'Alemberts principle yields

$$I\dot{\boldsymbol{\Omega}} = -R^2 M \dot{\boldsymbol{\Omega}}_{\parallel} - RMg\boldsymbol{n} \times \boldsymbol{e}_3 + \boldsymbol{\mu} \times \boldsymbol{B}.$$

Projecting, we obtain the tangential equation

$$I\dot{\boldsymbol{\Omega}}_{\parallel} = -R^2 M \dot{\boldsymbol{\Omega}}_{\parallel} - RMg\boldsymbol{n} \times \boldsymbol{e}_3 + \pi_{\parallel}(\boldsymbol{\mu} \times \boldsymbol{B})$$

and the normal equation

$$I\dot{\boldsymbol{\Omega}}_{\perp} = \pi_{\perp}(\boldsymbol{\mu} \times \boldsymbol{B}).$$

Again, we obtain the linear acceleration

$$\ddot{\boldsymbol{r}}(t) = R\dot{\boldsymbol{\Omega}} \times \boldsymbol{n} = \frac{RMg}{I + R^2 M} (\boldsymbol{e}_3 - \langle \boldsymbol{n}, \boldsymbol{e}_3 \rangle \boldsymbol{n}) + \frac{R}{I + R^2 M} \pi_{\parallel}(\boldsymbol{\mu} \times \boldsymbol{B}) \times \boldsymbol{n}.$$

The normal part of $\dot{\boldsymbol{\Omega}}$ does not contribute since its cross product with $\boldsymbol{n}$ vanishes.

We can directly observe multiple things:

1. The linear motion only depends on the tangential part of the magnetic torque. If the torque is a scalar multiple of $\boldsymbol{n}$, the center-of-mass motion will be just as in the non-magnetic case, but the rotation may still be more complicated.
2. The spin about the third axis is no longer conserved in general. It is only conserved if the torque is fully tangential to the plane.
3. Even if the torque is fully tangential, the equations of motion become considerably more difficult as long as $\boldsymbol{\mu}$ and $\boldsymbol{B}$ enter as they are time-dependent, so accelerations are no longer constant and we do not expect the sphere to behave like a point mass at the centre of gravity.

We do not expect the full equations to be analytically solvable since they are non-linear in the general case: The rotational matrices always introduce trigonometric terms via the torque. However, extreme cases may be of interest to understand the system. To consider this, we collect the full

equations for our system in particular in the magnetic case, denoting the torque by $\boldsymbol{\tau} := \boldsymbol{\mu} \times \mathbf{B}$ with tangential part $\boldsymbol{\tau}_{\parallel}$ and normal part $\boldsymbol{\tau}_{\perp}$:

$$\left\{ \begin{array}{l} \ddot{\mathbf{r}} = \frac{5g}{7}(0, \sin(\varphi) \cos(\varphi), 1 - \cos(\varphi))^{\top} + \frac{5 \cos(\varphi)}{7RM}(\boldsymbol{\tau}_{\parallel} \times (0, -\tan(\varphi), 1)^{\top}) \\ \dot{\boldsymbol{\Omega}}_{\parallel} = -\frac{5g \sin(\varphi)}{7} \mathbf{e}_1 + \frac{5}{2RM} \boldsymbol{\tau}_{\parallel} \\ \dot{\boldsymbol{\Omega}}_{\perp} = \frac{5}{2R^2 M} \boldsymbol{\tau}_{\perp} \\ \boldsymbol{\omega} = P^{\top}(t) \boldsymbol{\Omega} \\ \dot{P}(t) = \hat{\boldsymbol{\Omega}}(P(t)) \end{array} \right\}.$$

The Normal Torque The first special case is concerned with a fully normal torque $\boldsymbol{\tau} = \boldsymbol{\tau}_{\perp} \mathbf{n}$. The parallel part vanishes, the centre-of-mass motion reduces to the non-magnetic case, as does the tangential rotation. The differential equations for $\ddot{\mathbf{r}}$ and $\dot{\boldsymbol{\Omega}}_{\parallel}$ remain analytically solvable, but we have to take a closer look at $\dot{\boldsymbol{\Omega}}_{\perp}$. In the inertial frame, we have

$$\boldsymbol{\tau}(t) = (P(t)\boldsymbol{\mu}) \times \mathbf{B}.$$

The normal component is given by

$$\boldsymbol{\tau}_{\perp} = \langle ((P(t)\boldsymbol{\mu}) \times \mathbf{B}), \mathbf{n} \rangle.$$

For arbitrary $P \in \text{SO}(3)$, this equation remains highly non-linear, so we still depend on numerical solutions.

The Tangential Torque We turn our attention to the opposite extreme: $\boldsymbol{\tau} = \boldsymbol{\tau}_{\parallel}$. This tells us that $\dot{\boldsymbol{\Omega}}_{\perp} = 0$, so spin around the normal remains conserved. However, $\ddot{\mathbf{r}}$ and $\dot{\boldsymbol{\Omega}}_{\parallel}$ are now determined by non-linear ODEs and therefore likely not solvable.

Note to myself: The special cases are still too complicated to yield something analytical.

Minimal Initial Energy We assume for simplicity that $\mathbf{B} = \mathbf{e}_3$, ignoring units for the time being. If we allow the sphere to relax ideally before letting it roll, we can assume $\boldsymbol{\mu}(0) = \mathbf{e}_3$. This yields

$$\boldsymbol{\tau}(t) = (P(t)\mathbf{e}_3) \times \mathbf{e}_3 = (s_{\theta} c_{\psi}, s_{\psi} s_{\theta}, c_{\theta})^{\top} \times \mathbf{e}_3 = (s_{\psi} s_{\theta}, -s_{\theta} c_{\psi}, 0)^{\top}.$$

The projections are

$$\boldsymbol{\tau}_{\perp}(t) = \langle \boldsymbol{\tau}, \mathbf{n} \rangle = -\frac{\alpha}{\sqrt{\alpha^2 + 1}} \sin \theta(t) \cos \psi(t) \rightarrow \boldsymbol{\tau}_{\perp}(t) = \sin \varphi \sin \theta(t) \cos \psi(t)$$

and

$$\boldsymbol{\tau}_{\parallel}(t) = \boldsymbol{\tau} - \boldsymbol{\tau}_{\perp} \mathbf{n} = \begin{pmatrix} \sin \psi(t) \sin \theta(t) \\ -\cos^2 \varphi \sin \theta(t) \cos \psi(t) \\ -\sin \varphi \cos \varphi \sin \theta(t) \cos \psi(t) \end{pmatrix}.$$

Flat Surface Consider $\varphi = 0$ with $\mathbf{B} = B\mathbf{e}_3$ and $\boldsymbol{\mu}|_{t=0} = \mu(\mathbf{e}_2 + \mathbf{e}_3)$. Assume that the ball is initially at rest. We have $\mathbf{n} = \mathbf{e}_3$, so the motion is not influenced by gravity. Since the magnetic torque is fully directed in $\mathbf{e}_1$ -direction, we can parametrize $\boldsymbol{\mu}(t) = \mu(0, \sin \beta(t), \cos \beta(t))^{\top}$ with the angle β between $\boldsymbol{\mu}$ and $\mathbf{e}_3$. Plugging everything in the equations of motion, we obtain

$$\ddot{r}_2(t) = -\frac{R\mu B}{I + R^2 M} \sin \beta(t).$$

2.2. THREE-DIMENSIONAL SYSTEM

Assuming that $\beta(0)$ is small enough, we write $\sin\beta(t) \approx \beta(t)$. In addition, the rolling constraint yields $\dot{r}_2 = R\dot{\beta}(t)$, and we obtain

$$\ddot{\beta}(t) + \frac{\mu B}{I + R^2 M} \beta(t) = 0.$$

This is a second order linear ODE with analytic solution

$$\beta(t) = \beta_0 \cos(\omega t),$$

where $\omega^2 = \frac{\mu B}{I + R^2 M}$. The period of oscillation is $T = \frac{1}{2\pi} \sqrt{\frac{I + R^2 M}{\mu B}}$. We can use the rolling constraint to obtain the equation of motion

$$y(t) = y_0 + R(\beta(t) - \beta_0) = y_0 + R\beta_0(\cos(\omega t) - 1).$$

2.2.4 Non-Holonomic Mechanics: Damping

Up until now, the magnetic system is not subject to any dissipative forces. Especially the exclusion of friction is a simplification too severe to make quantitative predictions. We start by introducing rolling resistance. The frictional force is proportional to the modulus $N := Mg\langle \mathbf{n}, \mathbf{e}_3 \rangle$ of the surface normal force with proportionality constant ν_r . It acts opposed to the direction of motion, hence we can write

$$\mathbf{f}_r := \begin{cases} -\nu_r N \frac{\mathbf{v}}{\|\mathbf{v}\|} & \mathbf{v} \neq \mathbf{0} \\ \mathbf{0} & \mathbf{v} = \mathbf{0} \end{cases}$$

with $\mathbf{v} := \dot{\mathbf{r}}$. The new force is an angular force, so we adapt

$$\mathbf{F}_\theta = -I\dot{\boldsymbol{\Omega}} + \boldsymbol{\mu} \times \mathbf{B} - R\mathbf{n} \times \mathbf{f}_r$$

and obtain

$$I\dot{\boldsymbol{\Omega}} = -R^2 M \dot{\boldsymbol{\Omega}}_{\parallel} - RMg\mathbf{n} \times \mathbf{e}_3 + \boldsymbol{\mu} \times \mathbf{B} - R\mathbf{n} \times \mathbf{f}_r.$$

This yields analogously the equations of motions.

Flat Surface with Friction We reconsider the flat surface system with $\varphi = 0$, $\boldsymbol{\mu} = (0, \mu_2 \sin\beta(t), \mu_3 \cos\beta(t))^T$, and $\mathbf{B} = B_3 \mathbf{e}_3$. Assume no initial velocity. We can relate the velocity vector directly with $\dot{\beta}$ as $v_i = \delta_{i2} R \dot{\beta}$. We obtain

$$\dot{\boldsymbol{\Omega}} = (0, \dot{\beta}(t), 0)^T = \left[\frac{\mu B}{I + R^2 M} \sin\beta(t) - \frac{R^2 \nu_r M g}{I + R^2 M} \operatorname{sgn}(\dot{\beta}(t)) \right] \mathbf{e}_2$$

and hence

$$\ddot{\beta}(t) + \frac{\mu B}{I + R^2 M} \sin\beta(t) - R^2 \nu_r M g \operatorname{sgn}\dot{\beta}(t) = 0.$$

To obtain the canonical form of the ODE, define the undamped angular frequency $\omega_0 := \sqrt{\frac{\mu B}{I + R^2 M}}$ and the damping ratio $\zeta := \frac{R^2 \nu_r M g}{2(I + R^2 M)\omega_0}$, yielding

$$\ddot{\beta}(t) + \omega_0^2 \sin\beta(t) + 2\zeta\omega_0 \operatorname{sgn}\dot{\beta}(t) = 0.$$

Note that for the sphere, we have nicer expressions:

$$\omega_0 = \sqrt{\frac{5\mu B}{7R^2 M}}$$

and

$$\zeta = \frac{5\nu_r g}{14\omega_0}.$$

We again assume $\sin\beta(t) \approx \beta(t)$. The discontinuity at $\dot{\beta}(t) = 0$ is a numerical problem. Analytically, we can treat the cases $\dot{\beta} < 0$ and $\dot{\beta} > 0$ separately, for which the equation is a simple harmonic oscillator with dry friction, solved by

$$\beta_{\pm}(t) = A_{\pm} \cos(\omega_0 t) - (\pm)2\omega_0\zeta, \quad (2.13)$$

where $A_{\pm} = \beta_0 \pm 2\omega_0\zeta$ depending on the sign of $\dot{\beta}(t)$. At $\dot{\beta} = 0$, the two solutions are joined continuously. Note the difference to the viscously damped oscillator: Assuming $\beta(0) = \beta_0$ and $\dot{\beta}(0) = 0$, but $\dot{\beta}(t) < 0$ shortly after $t = 0$, we obtain the piecewise solution

$$\beta_{-}(t) = (\beta_0 + 2\omega_0\zeta) \cos(\omega_0 t) + 2\omega_0\zeta.$$

This holds up until the first turning point at $t_1 = \frac{\pi}{\omega_0}$, where it assumes the value $a_1 = 4\omega_0\zeta - \beta_0$. There, the solution's sign switches, and we obtain the amplitude $a_2 = \beta_0 - 4\omega_0\zeta$. This continues recursively, yielding

$$a_n = a_0 - n2\omega_0\zeta$$

for the amplitude at the n -th turning point. *Henceforth, the dry-damped oscillator has a linear envelope, not an exponential one.* Numerically, it may be desirable to avoid a jump discontinuity. Rather, one could use a smooth approximation, e.g. $\text{sgn}(\beta) \approx \tanh(\lambda x)$, where $\lambda > 0$ is a coefficient determining how rapid $\tanh$ converges to sgn away from $\dot{\beta} = 0$. This smooth interpolation is, however, not ideal. If λ is chosen too small or $\dot{\beta}$ approaches 0 very closely, the small angle approximation yields $\tanh(\lambda\dot{\beta}) \approx \lambda\dot{\beta}$, so we obtain

$$\ddot{\beta}(t) + 2\omega_0\bar{\zeta}\dot{\beta}(t) + \omega_0^2\beta(t) = 0,$$

where $\bar{\zeta} = \lambda\zeta$. This is the classical viscously damped harmonic oscillator, displaying the three different behaviours depending on $\bar{\zeta}$:

- If $0 < \bar{\zeta} < 1$, we have the underdamped case with solution

$$\beta(t) = \beta_0 \exp(-\bar{\zeta}\omega_0 t) \left(\cos(\omega_d t) + \frac{\bar{\zeta}\omega_0}{\omega_d} \sin(\omega_d t) \right), \quad (2.14)$$

where $\omega_d := \omega_0 \sqrt{1 - \bar{\zeta}^2}$.

- If $\bar{\zeta} = 0$, the system is critically damped with solution

$$\beta(t) = \beta_0(1 + \omega_0 t) \exp(-\omega_0 t). \quad (2.15)$$

- If $\bar{\zeta} > 1$, the system is overdamped with solution

$$\beta(t) = \frac{\beta_0}{\lambda_1 - \lambda_2} (-\lambda_2 \exp(\lambda_1 t) + \lambda_1 \exp(\lambda_2 t)) \quad (2.16)$$

for $\lambda_{1,2} := -\bar{\zeta}\omega_0 \pm \omega_0 \sqrt{\bar{\zeta}^2 - 1}$.

The underdamped case displays the characteristic exponential envelope, which can be used to judge the quality of the $\tanh$ approximation of sgn . We expect a transition between the linear and exponential envelopes numerically as $\dot{\beta}$ gets increasingly smaller.

CHAPTER THREE

Numerical Simulation

3.1 Numerics in C++

We begin by defining the exponential $\exp : \mathfrak{so}(3) \cong \mathbb{R}^3 \rightarrow \text{SO}(3)$, which takes a vector in $\mathbb{R}^3$ describing a rotation and maps it to the corresponding unit quaternion:

```
static quat exponential(const vec3 &rotvec){
    double theta = rotvec.norm();
    if (theta < 1e-15) return quat::Identity();
    vec3 axis = rotvec / theta;
    double a = cos(0.5 * theta);
    double b = sin(0.5 * theta);
    return quat(a, b * axis.x(), b * axis.y(), b * axis.z());
}
```

Furthermore, we need to be able to rotate a vector by a quaternion, which is already implemented in Eigen as follows:

```
static vec3 rotate(const quat &q, const vec3 &v){
    return q * v;
}
```

To calculate the torque, we first rotate the magnetic moment μ fixed to the body frame to the current position in the inertial frame and form the cross product:

```
static vec3 torque(const quat &q, const vec3 &mu, const vec3 &B){
    vec3 mu_init = rotate(q, mu);
    return mu_init.cross(B);
}
```

Using this, we can calculate $\dot{\Omega}$ by splitting into parallel and normal part from given parameters using the derived equation:

```
static vec3 getOmega(const vec3 &Omega, const quat &q, const
    Parameters &p){
    const vec3 e3(0.,0.,1.);
    vec3 tau = torque(q, p.mu_body, p.B_inert);
    vec3 dotOmega_t = - ((5 * p.g) / (7 * p.R)) * p.n.cross(e3) +
        (5 / (7 * p.R * p.R * p.M)) * (tau - (p.n.dot(tau)) * p.n);
}
```

```

    double dotOmega_n = (p.n.dot(tau)) / p.I;
    return (dotOmega_t + dotOmega_n * p.n);
}

static vec3 getr(const vec3 &Omega, const Parameters &p){
    return p.R * Omega.cross(p.n);
}

```

As one can see, $\dot{r}$ is easily obtained from $\dot{\Omega}$. The integrator itself uses a Runge-Kutta-Munthe-Kaas (RKMK) method as integrator. The general problem with using a naive RK approach is that it assumes a non-curved manifold, i.e. $\mathbb{R}^4$, to advance the functions. However, unit quaternions constitute $\mathbb{S}^3$, so an arbitrary step in $\mathbb{R}^4$ need not preserve the 3-sphere, eventually yielding a quaternion not representing any physical rotation. The RKMK approach uses the exponential to guarantee that one only advances by quaternionic multiplication. Since unit quaternions are a proper subgroup, they are closed under multiplication, so RKMK never diverges from the 3-sphere apart from precision errors. The idea is to advance $\dot{r}$ and $\dot{\Omega}$ by typical RK4 and use the exponential map to obtain the quaternionic steps. Therefore, $\Omega(t_{n+1})$ is obtained from $\Omega(t_n)$ by RK4, $r(t_{n+1})$ is obtained from $r(t_n)$, but q is obtained as

$$q(t_{n+1}) = \exp\left(\frac{\Omega(t_n) + \Omega(t_{n+1})}{2} dt\right).$$

Code-wise, this was implemented as

```

static State integrator(const State &s, double dt, const
    Parameters &p){
    vec3 Omega1 = s.Omega;
    //RKMK Steps
    quat q1 = s.q;
    vec3 k1_0 = getOmega(Omega1, q1, p);
    vec3 k1_r = getr(Omega1, p);
    vec3 Omega2 = s.Omega + 0.5 * dt * k1_0;
    quat q2 = exponential(0.5 * dt * Omega1) * s.q;
    vec3 k2_0 = getOmega(Omega2, q2, p);
    vec3 k2_r = getr(Omega2, p);
    vec3 Omega3 = s.Omega + 0.5 * dt * k2_0;
    quat q3 = exponential(0.5 * dt * Omega2) * s.q;
    vec3 k3_0 = getOmega(Omega3, q3, p);
    vec3 k3_r = getr(Omega3, p);
    vec3 Omega4 = s.Omega + dt * k3_0;
    quat q4 = exponential(dt * Omega3) * s.q;
    vec3 k4_0 = getOmega(Omega4, q4, p);
    vec3 k4_r = getr(Omega4, p);
    //RKMK Evaluation
    vec3 Omega_update = s.Omega + (dt / 6.0) * (k1_0 + 2.0 * k2_0 +
        2.0 * k3_0 + k4_0);
    vec3 r_update = s.r + (dt / 6.0) * (k1_r + 2.0 * k2_r + 2.0 *
        k3_r + k4_r);
    //Quaternion Update
    quat dq = exponential(dt * 0.5 * (s.Omega + Omega_update));
    quat q_update = dq * s.q;
    return State{r_update, Omega_update, q_update};
}

```

The accuracy of the integration w.r.t. the unit sphere can be measured by calculating the quaternion norm at each step. The algorithm sufficiently preserves energy if time is finely enough discretized, but it does not do so because of geometrical restraints (as opposed to the unit quaternion preservation).

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